A local moment refinement near the quartic zero-free boundary

A proposed analytic proof with an exact endpoint certificate

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11 October 2026 — Draft 3

Abstract

We propose a local refinement of the moment estimate in Collet's quartic zero-free argument. The coefficient of the total prime-slot length is reduced from 9/2 to 2249/500 = 4.498 in an explicit neighborhood of the relevant pair of ordinary summation lengths. Rows with a smaller buffered zero parameter are paid by removing slots and using the existing moment. Rows with a larger parameter are counted together with their arithmetic labels through one effective integer row. This draft supplies a proposed source-level analytic proof: exact signed characters and masks, the fixed character dictionary, the unmarked density input, restricted child norms, and the common Fourier transport are assembled in the supplements. An exact rational certificate supports the boundary 4374433/5000000 = 0.8748866. The final supplement gives a complete proposed written assembly of the weighted numerator and continuation argument for finite-order Eisenstein Hecke characters, ordinary Dirichlet characters, and the Riemann zeta function. This proof is under review and relies on the pinned incumbent analytic framework. It is not an independently verified record. Compiled Lean results support intermediate steps; they are not a complete formal proof of the proposed endpoint.

*Prepared with AI assistance in research, computation, formalization, and exposition. The upstream mathematical contributions are credited in Section 8. This draft has not received independent mathematical certification.

Contents

1 Introduction and the status of the result 4

2 The proposed local moment 5
2.1 Objects and normalization 5

3 Why a local improvement might close 6
3.1 The size of the budget defect 6
3.2 Rows with smaller zero parameters 7

4 Counting rows together with their labels 8
4.1 From sixth-power-free rows to all rows 8
4.2 A simple branch that explains the saving 8
4.3 The effective integer row in the general branch 9
4.4 What the proposed accounting gives 10

5 Other branches and uniformity 11
5.1 Comparison terms and additional masks 11
5.2 The amplifier errors 11
5.3 Exceptional terms cannot be hidden in an error 11

6 The exact conditional endpoint 12
6.1 The rational geometry and the exponent function 12
6.2 Why the local length restriction matters 13
6.3 The noncircular feedback mechanism 14

7 What is proved in writing and what is independently checked 14

8 Attribution and source relationship 15

A Reproduction and the certificate format 16

Guide to the supplementary source arguments 17

B Signed rows, zero extensions, and label reconstruction 17
B.1 The two signs at a common prime 17
B.2 The complete second frequency 17
B.3 Which labels are divisors of the row? 18

C The fixed dictionary behind the actual signed rows 19

D The unmarked plain moment and density for a fixed character dictionary 22

D.1 The zero-slot plain input 22
D.2 From actual witnesses to a height-uniform density bound 23
D.3 Formal verification scope 25

E Uniform deletion and common Fourier transport **25**

F Restricted canonical norms and the exact-width correction **27**
F.1 The actual coefficient and radial factorization 28
F.2 Payment before removing the restriction 28
F.3 Actual width versus nominal width 29

G Assembly of the local physical moment **30**

H Shortened physical scales do not require a new local hypothesis **36**

I The endpoint and the target-uniform continuation argument **40**

1 Introduction and the status of the result

A fixed zero-free half-plane is a statement that an L-function has no zeros to the right of a specified vertical line. For the families under discussion, lowering that line is an improvement. The eventual aim of this project is a strict half-plane $\Re s > \theta$ for ordinary Dirichlet L-functions, including the Riemann zeta function, and for finite-order Hecke L-functions of trivial infinite type over $\mathbf{Q}(\sqrt{-3})$. Principal poles are excluded. A bound near 0.875 is a fixed-strip improvement; it does not approach a proof of the Riemann hypothesis by itself.

Our starting point is the pinned Cycle25 manuscript of Collet [1], which builds on OpenAI's 7/8 framework [2] and Levy's weighted numerator [3]. Its exact comparison boundary is

$$b_0 = \frac{11}{12} - \frac{\ell_0}{4} = 0.8748866470399641\ldots, \qquad P(\ell_0) = 0, \tag{1}$$

where

$$P(X) = 927X^4 - 3135X^3 + 2433X^2 + 275X - 100, \qquad \ell_0 \in [1/6, 1/5].$$

Throughout this paper, "comparison boundary" refers to this particular pinned source, rather than to a claim about the current leaderboard.

The proposed modification is deliberately local. In the source proof, two ordinary character sums and a product of short prime sums share a moment budget. Write their logarithmic lengths as n_1, n_2, z and the row budget as M. The existing estimate is available with $n_1 + n_2 + (9/2)z \le M$. We seek to replace $9/2$ by 4.498 only when the lengths lie near the configuration used by the endpoint argument. Because $z > 0$, a smaller coefficient permits a slightly larger prime product at the same row budget. The weighted numerator can then gain a small amount.

The proposed endpoint is

$$\theta = \frac{4374433}{5000000} = 0.8748866. \tag{2}$$

Its distance below (1) is about 4.704×10^{-8}. The size of this gain makes exact arithmetic essential. More importantly, a favorable scalar inequality is useful only if it belongs to a valid estimate for the original arithmetic sums.

The status of each assertion matters. The elementary inequalities and rational endpoint certificate are established within their algebraic scope. The source-level analytic argument presented here is a proposed proof under review, relative to the pinned incumbent theorems. Several of its intermediate identities and estimates have compiled Lean proofs using cached dependencies. A complete new formal endpoint theorem and independent verification are absent. The all-family conclusion at (2) is the proposed theorem in Section I, with a complete written argument supplied for review. Its proof status must be distinguished from independent acceptance or formal certification.

Here is the roadmap. Section 2 states the exact family and local estimate. Sections 3–5 explain its two branches and the reserves paying the remaining terms. Section 6 gives the conditional numerical consequence. The supplements then present the detailed source arguments in dependency order: signed reconstruction, the fixed character dictionary, plain density, uniform Fourier transport, restricted canonical children, the assembled local moment, and shortened physical scales. The final supplement assembles the full numerator and all-family continuation, including short selected products and every free Euler shift. The independent-validation scope is stated in Section 7.

Input or conclusion	Role in the argument
Pinned 7/8 and 9/2 results	Existing analytic inputs; called on children with their actual masks and moving characters.
Signed row and fixed dictionary	Identify the character being counted and preserve literal eligibility.
Low-row payment and high-row density	The two parts of the proposed local 4.498 moment.
Exact rational certificate	Converts that moment, if valid in the full numerator, into a strict endpoint saving.
All-family endpoint	Proposed complete written proof relative to the pinned framework; independent review and complete formal verification remain outstanding.

2 The proposed local moment

2.1 Objects and normalization

We use the natural moment family of [1, Appendix B3.4] over $\mathcal{O} = \mathbf{Z}[\omega]$, where $\omega = e^{2\pi i/3}$. Write N for the ideal norm, or the norm of an element. Set $U = Z^M$, with $Z > 1$ and $M > 0$. At the parent stage the character acting on a summation variable n has the form

$$\psi_k(n) = \tau(n)\chi_n(k),$$

where τ is a fixed finite ray character and $\chi_n(k)$ is the source's sextic residue character, with its prescribed zero extension. The ray modulus is fixed independently of U. No additional parent character of positive moving width is allowed in our proposed estimate. Such factors can occur in transformed children and must still be paid.

Let $S_{k,1}(Z^{n_1})$ and $S_{k,2}(Z^{n_2})$ denote the source's centrally normalized ordinary character sums. The two smooth profiles, real norm powers, and height parameters may be independent. A prime slot has the form

$$Q_{k,i} = Z^{-z_i/2} \sum_{\mathfrak{p} \in \mathcal{P}_i^{\mathrm{can}}} \psi_k(\mathfrak{p})\nu_i(\mathfrak{p}) W\left(\frac{N\mathfrak{p}}{Z^{z_i}}\right) \left(\frac{N\mathfrak{p}}{Z^{z_i}}\right)^{-\sigma_i + iv_i}. \tag{3}$$

Here $\mathcal{P}_i^{\mathrm{can}}$ is the genuine canonical good-prime pool, ν_i is an allowed relative ray character, and the slots share the one smooth annular profile W. Put $z = \sum_i z_i$ and $Q_{k,I} = \prod_{i \in I} Q_{k,i}$. Formula (3) does not permit arbitrary prime subsets or independently chosen slot profiles.

An additional common puncture is allowed: all factors may be restricted to arguments coprime to a fixed nonzero mask of norm at most U^B. It stays fixed during the current row sum but may depend on labels already frozen in a preceding transformation. The underlying natural zeros must also stay in place. A canceled character phase at a nonunit does not turn the original value zero into one.

The word *eligible* means the literal source predicate excluding `FixedInducingRow` presentations. Let M_{ray} be the fixed ray modulus and η_0 the fixed anchor in the relative slot characters. The initial source uses

$$Q_* = M_{\mathrm{ray}} \cap (72) \cap \mathfrak{q}_{\eta_0}.$$

This ideal is held fixed in transformed states, even when their whole-product character moves. Its containments cover every allowed relative ray twist; they are part of the initial family, not a later change in the exceptional set. Eligibility is not replaced by nonprincipality of one preferred

primitive representative. Its exceptional part is treated by the centered cancellation. The exact transport is spelled out in Section C.

Proposition 2.1 (Local physical moment: proposed proof under review)**.** *Use the preceding family, with all its natural masks and canonical slot conditions. Suppose*

$$\frac{41M}{100} \le n_i \le \frac{11M}{25}, \qquad \frac{3M}{100} \le z \le \frac{M}{25}, \qquad n_1 + n_2 + Cz \le M, \qquad C = \frac{2249}{500}. \tag{4}$$

For every $\varepsilon > 0$, the proposed estimate is

$$\sum_{k \text{ eligible}} |S_{k,1}(Z^{n_1}) S_{k,2}(Z^{n_2}) Q_{k,I}|^2 \ll Z^{M+\varepsilon} \mathcal{H}_J. \tag{5}$$

The source's radial row weight is understood. The factor $\mathcal{H}_J$ denotes its finite smooth-seminorm and polynomial height allowance, retaining independent heights. The assertion also includes the four annular derivative-profile pairs and bounded scale-box recovery used by the physical numerator. All quantifiers, including selection of a common fine mesh before the later arithmetic data, are as in the source family.

Proof reference. The exact source-level proof is given in Section G, after the signed reconstruction, fixed-dictionary, plain-density, Fourier, and restricted-child lemmas. □

The notation imports a substantial existing analytic framework. This draft explains a modification of that framework; it is not a replacement construction of the ordinary L-functions or of the source moment family. Sections F and G provide the proposed connection to the actual canonical state and the complete local source assembly. They are written analytic arguments with compiled intermediate steps, rather than a new formal endpoint theorem.

3 Why a local improvement might close

There are two potential difficulties with improving a moment estimate. First, a transformed child need not remain in the same local length window as its parent. Second, counting a character family after freezing its arithmetic labels may be too expensive. The proposed argument treats these difficulties separately.

Apply the two transformations in the existing moment proof and classify the resulting nonexceptional rows by a buffered zero parameter δ. Roughly, the associated real zero location is $(1+\delta)/2$; the buffer and finite character family are those of the source. When $\delta \le 7/10$, pointwise prime estimates make it cheap to remove enough slots to recover the existing global moment condition. For $\delta \ge 7/10$, those pointwise estimates are more expensive, but the rows are sparse. The proposed saving comes from counting each such row *together with* its common-support labels.

The split is made after the transformations. No Poisson formula is applied to a zero-bin indicator. Also, the smaller-parameter branch calls the existing moment rather than calling Proposition 2.1 on a shortened child. Thus there is no claimed local induction whose length window must somehow survive indefinitely.

3.1 The size of the budget defect

Define

$$A = n_1 + n_2 + z, \qquad g = (n_1 + n_2 + \tfrac{9}{2}z - M)_+, \qquad a = \frac{M}{2000}, \qquad \sigma = 3a. \tag{6}$$

The auxiliary amplifier has prime norms on the scale Z^a, disjoint from the live slot windows after the mesh is chosen sufficiently fine. Its length a is fixed relative to M; it is not an arbitrarily small final error.

Lemma 3.1 (Elementary parent bounds)**.** *Under* (4)*,*

$$0 \le g \le \frac{M}{12500}, \qquad \frac{85M}{100} \le A \le \frac{44753M}{50000} < \frac{9M}{10}.$$

Proof. The difference of the two capacity coefficients is $9/2 - C = 1/500$. Hence $g \le z/500 \le M/12500$. The lower bound for A follows by adding the lower length bounds. For the upper bound, use

$$A \le M - (C-1)z \le M - \frac{1749}{500}\frac{3M}{100} = \frac{44753M}{50000}.$$

□

3.2 Rows with smaller zero parameters

This branch uses the old moment on the actual transformed state. Let M_{act} be that state's width: the logarithm of its dyadic row scale plus the logarithm of its displayed character modulus. Let M_{nom} be the nominal width in the exponent ledger. They need not be equal. For actual clipped plain lengths x_1, x_2 and remaining slot widths z_i, put

$$F_{\mathrm{act}} = (x_1 + x_2 + \tfrac{9}{2}\sum_i z_i - M_{\mathrm{act}})_+.$$

One greedy selection removes a total slot width at most $F_{\mathrm{act}}/(9/2) + \eta_{\mathrm{mesh}}$ until the old capacity condition holds or every slot has been removed. On the selected low rows, the buffered prime estimate charges $k_0 = 7/10$ per unit width in squared norm. Only after this payment is the low-row restriction dropped by positivity. The remaining norm is exactly the canonical norm to which the old moment applies. If no slot remains, its unrestricted zero-slot estimate has no further plain-capacity condition.

The comparison of actual and nominal widths is essential. Set $d = M_{\mathrm{act}} - M_{\mathrm{nom}}$, with $d \le \eta_2$, and suppose the nominal defect is bounded by $F_{\mathrm{nom}} \ge 0$. Then $F_{\mathrm{act}} \le (F_{\mathrm{nom}} - d)_+$. For $0 \le r \le 1$,

$$d + r(F_{\mathrm{nom}} - d)_+ \le \eta_2 + rF_{\mathrm{nom}} \qquad (\eta_2 \ge 0). \tag{7}$$

This follows by considering $F_{\mathrm{nom}} - d \ge 0$ and its complement. Thus a smaller actual width offsets its larger capacity defect. One must not discard this offset and claim that the actual defect itself is bounded by the nominal one.

For the main low branch, $r = k_0/(9/2) = 7/45$ and $F_{\mathrm{nom}} \le g + 6a + \eta$, where η includes the enlarged-ball and support padding. The exact source child saving is at least a. Therefore

$$a - \frac{k_0}{9/2}(6a + g) \ge \frac{M}{2000} - \frac{7}{45}\left(\frac{3M}{1000} + \frac{M}{10000}\right) = \frac{4M}{225000} > 0. \tag{8}$$

Choose the mesh, buffer and aggregate support losses within this fixed reserve. The actual coefficient factorization and radial restriction, including why no indicator is Poisson-summed, are proved in Section F. The compiled results check the restriction, the greedy choice, and inequality (7).

4 Counting rows together with their labels

4.1 From sixth-power-free rows to all rows

The unmarked density exponent supplied by the existing source is

$$\rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)}, \qquad \frac{7}{10} \le \delta \le \frac{3}{4}. \tag{9}$$

The adjective "unmarked" means that this count requires no positive prime-slot supply. It does not mean that arbitrary moving ray characters can be inserted. A single fixed finite character list is selected before the live labels.

Every nonzero row has a sixth-power decomposition $j = kb^6$, with the source's primary-generator and unit conventions. The extra factors of b leave Euler holes in the displayed character. In the positive half-plane they preserve the zeros of its primitive inducing character. This permits the following general counting step.

Lemma 4.1 (Summing the sixth-power part)**.** *Assume a uniform count $\ll Y^{\rho+\varepsilon}$ for the relevant sixth-power-free cores of norm at most Y, for every $Y \ge 1$. If $\rho \ge 2/7$, the corresponding count of all rows of norm at most $H \ge 1$ is $\ll H^{\rho+\varepsilon}$. The constant may be chosen uniformly for ρ in a fixed interval with lower endpoint $2/7$, provided the core estimate is itself uniform.*

Proof. Up to the fixed unit-sector convention, the count is at most

$$H^{\rho+\varepsilon} \sum_{Nb \le H^{1/6}} (Nb)^{-6(\rho+\varepsilon)}.$$

The ideal series converges since $6\rho \ge 12/7 > 1$, and a single convergent series with exponent $12/7$ dominates the sum uniformly. If the input estimate was initially only eventual, bounded windows must first be paid with a uniform constant. This is a genuine requirement, not an automatic consequence of an eventual estimate with varying data. □

Here $\rho(7/10) = 19/54$ and $\rho(3/4) = 2/7$, and ρ is decreasing. Fixed-inducing presentations stay excluded from this count; their exceptional contribution must be treated separately.

4.2 A simple branch that explains the saving

Consider the second-transform branch in which the same squarefree label P occurs once on each side, with $\log_Z NP = q$. Its actual character row is

$$j = P^2 h', \qquad \log_Z Nj \le M - 6a + o(1).$$

At a fixed j, the labels P satisfy $P^2 \mid j$. There are only divisor-many such labels, at an arbitrarily small power cost. Thus a density bound on j controls the joint population (P, h') without an additional factor Z^q.

This ordering matters. If one first freezes P, bounds the remaining rows for each twist, and finally pays for all choices of P, one loses the information that P is already contained in j.

For this branch the source normalization leads to the proposed exponent

$$A - M + 5a + (M - 6a)\rho(\delta) + \delta(A - q). \tag{10}$$

Relative to the allowance A, it has excess $M(\rho-1) + \delta A + a(5-6\rho) - \delta q$. Equation (10) illustrates the joint count. The general identity below, including its masks, supplies the arithmetic statement needed beyond this illustrative branch.

4.3 The effective integer row in the general branch

The first common-support extraction introduces widths c, d, p, where p is now the first common radical width. Let R be its active conductor width and E the mask-divisor width. These are local bookkeeping variables; the p of Section 6 will instead denote a detector parameter. Set $T = (c-d)_+$.

The signs of the two separated twists are opposite. We fix the convention of the formal source: $t_{\mathfrak{p}}$ is the left multiplicity minus the right multiplicity, modulo six. The first coefficient contains the active character on the left and its conjugate on the right. After reciprocity, the exact row residues are therefore

$$e_C(\mathfrak{p}) = (1 + t_{\mathfrak{p}}) \bmod 6, \qquad e_D(\mathfrak{p}) = (1 - t_{\mathfrak{p}}) \bmod 6.$$

To see the sign directly, write $c_n(\pi)$ for the zero-extended sextic row and $q(\pi, n)$ for its fixed reciprocity phase. Reciprocity and multiplicativity give

$$c_n(\pi)c_\pi(n)^t = q(\pi, n)^t c_n(\pi)^{1+t}, \qquad c_n(\pi)\overline{c_\pi(n)^t} = q(\pi, n)^{6-t} c_n(\pi)^{7-t}.$$

For $1 \le t \le 5$, these are identities including nonunit zeros. An exponent six is the coprimality mask, not the constant one. Section B gives the full conventions and explains the unit relating an ideal generator to chosen prime generators. Define

$$F_C = \prod_{\mathfrak{p}|r} \mathfrak{p}^{e_C(\mathfrak{p})}, \quad L_C = \log_Z NF_C, \quad R_{\mathrm{zero}} = \sum_{\substack{\mathfrak{p}|r \\ e_C(\mathfrak{p})=0}} \log_Z N\mathfrak{p}.$$

An exponent zero here records cancellation of a phase; its old puncture is still retained.

The signed identity after the second common-support extraction gives

$$w = F_C e G_c V_{\mathrm{id}} h', \qquad \log_Z Nw \le M - R - T + L_C - 6a + o(1). \tag{11}$$

The factors e, G_c, V_{id} are the source's mask and second extraction factors. The analytic row ball must actually be enlarged to the smooth envelope

$$H = K_0 + (d - c)_+ + 6a + \varpi, \qquad K_0 = 2A - c - d + R + E - M, \tag{12}$$

before the second transformation. Here $\varpi > 0$ pays the retained frequency padding and the fixed support-doubling factor. If the retained first dyad satisfies $K \le K_0 + \delta_1$, take $\varpi \ge \delta_1$ with a further positive support reserve. The amplified rows have norm at most Z^{K+6a}, so the actual smooth ball contains them. This is exactly the source's `mainCommonRadius` with $\sigma = 3a$. The added ϖ only shortens the dual range in (11); its low-branch cost was reserved above. For a fixed parent character modulus, write $M = m + q$ and use m in the formula for K_0; the same final total-width identity results. In particular, the actual nominal child width is

$$M_{\mathrm{nom}} = M - (c - d)_+ - 6a - \varpi - g_2 + t_2 \le M - 6a - \varpi.$$

At fixed w, the reconstruction makes the active first primes, the mask primes, the second common primes, and the first residual projector label divisor-many. First common primes absent from w must still be counted; their width is

$$p - R - E + R_{\mathrm{zero}}. \tag{13}$$

In particular, a canceled prime cannot simultaneously be omitted from w and treated as one of its divisors. A general arithmetic justification of this counting step is now given in the companion

reconstruction note [5]. At a fixed row w, write L for the radical of first common primes absent from w. All the remaining labels are supported on $\mathrm{rad}(w)L$. For a fixed bound on their logarithmic norms, Rankin's inequality gives, for every $\varepsilon > 0$,

$$\#\{I : NI \le Z^T,\ \mathrm{rad}\, I \mid Q\} \ll_{B,T,\varepsilon} Z^{\varepsilon} \qquad (NQ \le Z^B).$$

Indeed the count is at most $Z^{aT}\prod_{\mathfrak{p}\mid Q}(1-(N\mathfrak{p})^{-a})^{-1}$; take $aT \le \varepsilon/2$ and bound the product by a constant times $(NQ)^{\varepsilon/(2B)}$. The finite exceptional set of small primes contributes only a constant. Counting the missing radical costs its norm exponent (13). This is a general reconstruction bound, rather than a finite sample of valuations. The source identities supplying its factorization and mask assumptions are given in Section B. The second residual projector is different from the first one: it need not divide w. Keep it fixed outside the row sum, retain its full signed allocation, and pay its count together with both extraction scalars by B3.67–B3.69. No extra puncture is imposed on the allocated quotients.

The first labels and the full coprimality projector remain inside the positive norm until this joint count is applied. Their Möbius signs are retained through the exact allocation. A common Fourier measure is chosen on the ambient parameter box before labels and bins; only its evaluation points depend on the later arithmetic data. This order is needed to keep independent heights and smooth-profile bounds uniform.

4.4 What the proposed accounting gives

Let b_2, g_2, t_2 denote the second common widths, with $0 \le g_2 \le b_2$ and $t_2 \ge 0$, and put $B_c = (3c - 5d - R)_+/6$. Relative to the joint allowance $p + A - c + R + E + B_c$, the proposed excess is

$$\begin{aligned}\mathcal{E} = {} & M(\rho-1) + \delta A + a(5-6\rho) \\ & + (1-\rho)T - \delta c - (1+\rho)R - E - B_c + \rho L_C + R_{\mathrm{zero}} \\ & + g_2 - t_2 - (1+\delta)b_2 + \eta. \end{aligned} \tag{14}$$

The rational audit checks this algebraic ledger. Its analytic input is the signed reconstruction and exact extraction normalization, followed by the fixed-dictionary density estimate and common Fourier transport in Sections B–E. Section G gives their proposed composition.

The favorable signs can be checked directly. Using $c = \min(c,d) + T$, $\min(c,d) \ge p$, $R \le p$, and $L_C \le 5(R - R_{\mathrm{zero}})$, the second line of (14) is at most

$$(4\rho - 1 - \delta)p - E - B_c \le -\frac{79}{270}p - E - B_c.$$

Indeed $\rho + \delta - 1 > 0$, $5\rho - 1 > 0$, and $4\rho - 1 > 0$ throughout the interval. Also $4\rho - 1 - \delta$ is decreasing and equals $-79/270$ at $\delta = 7/10$. The third line without η is at most $-\delta b_2 - t_2 \le 0$.

Finally (4) implies the weaker bound $A/M \le 1 - (16/5)(3/100) = 113/125$. Since

$$\rho'(\delta) = -\frac{50}{3(5-2\delta)^2},$$

the function $\rho(\delta) + (113/125)\delta$ decreases on $[7/10, 3/4]$. Therefore

$$1 - \rho(\delta) - \delta A/M \ge \frac{259}{16875}, \qquad 5 - 6\rho(\delta) \le \frac{23}{7}. \tag{15}$$

For $a = M/2000$ and $\eta \le M/1000$, equations (14)–(15) give

$$\mathcal{E} \le -\frac{259M}{16875} + \frac{23M}{14000} + \frac{M}{1000} < -\frac{M}{100}.$$

This leaves a substantial reserve for support and buffering losses. The distinction between the scalar verification and its proposed analytic source assembly is retained in Section 7.

5 Other branches and uniformity

5.1 Comparison terms and additional masks

The centered source uses equal-product comparison rectangles with one plain length $M/4$. For an additional common mask, the proposed argument expands that mask only while bounding the comparison terms. After reflecting the second natural daughter if its length exceeds $M/2$, the two plain lengths sum to at most $3M/4 + \xi$. Their existing-moment budget is consequently at most

$$\frac{3M}{4} + \frac{9}{2}\frac{M}{25} + \xi = \frac{93M}{100} + \xi < M$$

for sufficiently small padding ξ. This pays comparison daughters without asserting that they belong to the local window (4). The mask remains on both rectangles of the original centered expression.

5.2 The amplifier errors

For the three extracted valuation alternatives $1, 6, 7$, the relevant width pairs are (λ, λ), $(6\lambda, 0)$, and $(7\lambda, \lambda)$, with $\lambda \ge \sigma/4$. The old moment pays these branches. The source's defect estimate is

$$F_{\mathrm{nom}} \le g + \sigma + w - w_o + \eta,$$

its combined width and removal cost is at most $\eta_2 + F_{\mathrm{nom}}/6$ by (7), and its saving is at least w. Substituting $g \le M/10000$ gives the following lower margins before padding:

Valuation	(w, w_o)	Saving minus base cost
1	(λ, λ)	$13M/120000$
6	$(6\lambda, 0)$	$193M/120000$
7	$(7\lambda, \lambda)$	$119M/60000$

The normalized prime average must be retained. Introducing a second count of the auxiliary primes would change these margins.

5.3 Exceptional terms cannot be hidden in an error

Lemma 3.1 gives $A - M < -M/10$. The source's second zero-frequency term has excess at most $A - M + 5\sigma/3 + \eta$, and the centered fixed-inducing calculation retains a maximum $A - M$ before its terminal loss 3σ. Both fit within this negative margin. Replacing $A - M$ by $(A - M)_+$ would discard the saving that pays the fixed amplifier.

For the enlarged row ball (12), the first exceptional reserve is checked at that actual radius. Before the ϖ padding, the exact expressions are

$$F_1 = \begin{cases} (c + 5d + R + E)/6 + B_c - 4a, & c \geq d, \\ c + (R + E)/6 + B_c - 4a, & c \leq d. \end{cases}$$

The padding subtracts $5\varpi/6$. Thus $F_1 \geq 2c/3 - 4a - 5\varpi/6$, while $F_2 \geq 2b_2/3$. The earlier $-5a$ expression was a weaker lower bound, not an equality: it dropped the positive amplifier-density term a. The exact centered cancellation, forced congruences and raw extraction factors are retained in Section G. The resulting main exceptional loss $4a + O(\eta)$ fits within $M - A$; the original error branches retain their $3\sigma + O(\eta)$ bound.

Uniformity is part of the assertion. One first fixes the requested final loss, then chooses the aggregate small losses below both that loss and the margin in (8). Profile orders, height orders, and the eventual threshold are chosen through the finite chain of transformations. A loss per slot, or a character family chosen only after the live labels, would not supply the required estimate.

6 The exact conditional endpoint

This section describes an independently reproducible scalar conclusion. It uses the existing detector function; only the physical moment coefficient is changed. Here $p \in [0, 1/2]$ is a detector parameter, unrelated to the common radical width in Section 4.

6.1 The rational geometry and the exponent function

Set

$$\begin{aligned} \ell &= \frac{626701}{3750000}, & h &= \frac{3969737987}{5000000000}, \\ x &= 1 + \ell - h = \frac{5597590039}{15000000000}, & y &= h - 2\ell = \frac{6895605961}{15000000000}, \\ \kappa_* &= 2\theta - 1 = \frac{1874433}{2500000}, & C &= \frac{2249}{500}. \end{aligned} \tag{16}$$

Then $\theta = 11/12 - \ell/4$ and $x + y = 1 - \ell$. For $\kappa \geq \kappa_*$ define

$$\begin{aligned} D_\kappa(p) &= 3 - p - \frac{2p}{3\kappa}, & B_\kappa(p) &= 2(1-p)\left(1 - \frac{p}{3\kappa}\right), \\ c_\kappa(p) &= B_\kappa(p)/D_\kappa(p), & V &= \frac{5}{6} - \delta, \\ R_\kappa(\delta, p) &= 1 - \delta + \frac{\delta c_\kappa(p) V}{2(V + \delta c_\kappa(p))}. \end{aligned} \tag{17}$$

We use a lower-case c_κ to distinguish the detector coefficient from the physical coefficient C. The candidate upper-row exponent is

$$F_{C,\kappa}(\delta, p) = \frac{1+\ell}{2} + \frac{\delta}{2} + h\left(\frac{3R_\kappa(\delta,p)}{4} - \frac{5}{12}\right) + \delta p\left(\ell - \frac{h - 4\ell}{2C}\right). \tag{18}$$

Proposition 6.1 (Exact scalar certificate)**.** *With* (16)*,*

$$F_{C,\kappa_*}(\delta, p) \leq \theta - 10^{-7} \quad \textit{for every } (\delta, p) \in [0, 3/4] \times [0, 1/2].$$

Moreover $\theta < b_0$ *for the exact boundary* (1)*.*

Computer-assisted proof of the first assertion. All constants in this calculation are rational. Put $J = VD_{\kappa_*} + \delta B_{\kappa_*}$. Then

$$R_{\kappa_*} = \frac{(1-\delta)J + \delta B_{\kappa_*} V/2}{J}.$$

The stored certificate first gives $J \geq 1/10$ on the whole rectangle. After multiplying $\theta - 10^{-7} - F_{C,\kappa_*}$ by J, it suffices to verify a polynomial inequality. The audit partitions the rectangle into 18 rational subrectangles and writes the polynomial in the Bernstein basis on each one. Every coefficient is nonnegative; the recorded closed subrectangles cover the entire original domain. Appendix A identifies the executable audit and its exact certificate. This is an exact finite verification, not a sampling of the exponent on a numerical grid. □

Proof of the comparison with the quartic. We have $P(1/6) < 0$, and exact substitution gives

$$P(\ell) = \frac{3484615068030068970503}{219726562500000000000000000} > 0.$$

For $X \in [1/6, 1/5]$,

$$P'(X) = 3708X^3 - 9405X^2 + 4866X + 275 \geq -\frac{9405}{25} + \frac{4866}{6} + 275 = \frac{3549}{5} > 0.$$

Thus $\ell > \ell_0$, and $\theta = 11/12 - \ell/4 < b_0$. □

6.2 Why the local length restriction matters

The row exponent in the numerator varies; it is not always h. Let d denote that exponent and t a normalized free Euler shift. The local estimate is intended only for

$$h - 10^{-3} \leq d \leq h + 10^{-10}, \qquad 0 \leq t \leq 10^{-3}. \tag{19}$$

The audit checks that both reflected plain lengths, normalized by the row budget, lie strictly between 0.41 and 0.44, and that the selected prime width lies strictly between 0.03 and 0.04. It also checks that the old 9/2 defect is less than 1/12500 and that the original physical slots supply the selected product.

For clarity, a general scalar row exponent used by the audit is

$$\begin{aligned} G_{c,\kappa}(d;\delta,p) = \frac{1}{2} + h\left(\frac{17}{50} - \frac{1}{6}\right) - \frac{y}{2} - \frac{\ell}{2} + d\left(\frac{1}{4} - \frac{17}{50}\right) + \frac{\delta}{2} \\ + \frac{3d}{4} R_\kappa(\delta,p) + \delta p\left(\ell - \frac{2y-d}{2c}\right). \end{aligned} \tag{20}$$

It is affine in d on the range in question, since $d \geq 1/2 > y$. At $d = h$ and $c = C$ it reduces to (18). The exact checks are

Row endpoint	Coefficient c	Saving below θ	Rectangles
h	C	10^{-7}	18
$h - 10^{-3}$	$6\kappa_*$	10^{-5}	6
$1/2$	$6\kappa_*$	10^{-3}	1

The lower interval therefore uses the old estimate. At the junction, $C < 6\kappa_*$ and $2y - d > 0$, so the local exponent is no larger than the old one. Affinity bounds the intermediate local rows. Above h, a scalar slope bound of 2 leaves saving $10^{-7} - 2 \cdot 10^{-10} > 9 \cdot 10^{-8}$.

For larger free shifts the source's existing negative coefficient contribution must be kept. Write $q = \delta p$, so $0 \le q \le \delta/2$. The audit verifies, for the normalized available width in this window,

$$-\frac{\delta t}{2} - \frac{q z_{\mathrm{old}}(t)}{2} \le -\frac{q z_{\mathrm{new}}(0)}{2}, \qquad t \ge 10^{-3}.$$

Here $z_{\mathrm{old}}(t) = \max(0, b-2t)/c$ with $6\kappa_* \le c \le 9/2$, and $z_{\mathrm{new}}(0) = b/C$; b is the residual budget after the unshifted reflected plain factors. The normalized reserve at the first shift exceeds $3/10000$ and increases thereafter, including after the old available width becomes zero. The free-shift comparisons are exact scalar interface checks. Section H explains why downward changes of the reflected plain scales can use either the local moment or the old moment without changing the weighted coefficient mass. If the selected physical prime product is shorter than the local lower bound, the old moment has reserve $1 - 2(211/500) - (9/2)(3/100) = 21/1000$. The slot-selection theorem still measures its gain at the nominal target capacity. Section I proves this distinction and covers every physical Euler term, including an empty selected product.

6.3 The noncircular feedback mechanism

Section I inserts Proposition 2.1 into the original weighted numerator with its coefficients, masks, conductor losses, principal terms, and family reductions preserved. The numerical feedback used in that proof is as follows. Let β_* be the supremum used in the source, with its sentinel $1/2$. The $7/8$ initialization gives $\beta_* \le 7/8$. If $\beta_* > \theta$, put $\kappa = 2\beta_* - 1$; then $\kappa_* < \kappa \le 3/4$.

For the new upper-row expression, only the detector depends on κ. The source bounds $\partial_\kappa c_\kappa < 1/18$ and $\partial_c R \le 5/12$ give

$$\partial_\kappa F_{C,\kappa} \le \frac{3h}{4}\frac{5}{12}\frac{1}{18} = \frac{3969737987}{288000000000} < \frac{1}{50}.$$

Consequently Proposition 6.1 implies

$$F_{C,2\beta_*-1}(\delta, p) - \beta_* \le -10^{-7} - \frac{24}{25}(\beta_* - \theta) < 0. \tag{21}$$

For old rows the coefficient 6κ also varies. The separate derivative estimate in Section I gives $\partial_\kappa G_o < 1/8$, leaving at least $3(\beta_* - \theta)/4$ in addition to their certificate savings. Together these strict savings feed the source's contradiction to the definition of β_*. The hypothesis on the slot zero bound is an equality at the chosen κ; the improved zero-free conclusion is not used as an input. The final supplement supplies the analytic numerator and continuation steps that turn this feedback into the proposed conclusion $\beta_* \le \theta$.

7 What is proved in writing and what is independently checked

Draft 3 supplies a complete proposed written analytic argument for the local moment and the all-family endpoint, relative to the pinned source results named in its supplements. It addresses the questions left open in Draft 2: signed characters and zero extensions, the fixed dictionary and unmarked plain moment, uniform height thresholds, canonical low-row restriction, enlarged radii,

and the full local branch ledger. The final assembly includes shortened plain factors, a short or empty selected prime product, free Euler shifts, principal and tail terms, continuation, and the passage to ordinary Dirichlet and zeta functions. These arguments are supplied for review; their inclusion does not create a verified tracker record.

The established finite evidence has a narrower scope. The rational audits cover continuous sign inequalities, exact symbolic exponents, and complete rational Bernstein certificates. Finite CRT and joint-fiber checks help detect mistakes but are not used in place of the general reconstruction proof. The current development replay compiles 15 components and an audit module, with 48 printed theorem axiom checks. They use cached pinned dependencies, so this is not a fresh rebuild of the entire dependency graph.

Evidence group	What it establishes
Finite deletion and sixth powers	Preservation of natural holes and positive-half-plane zero data.
Buffered slots and restricted incumbent	Pointwise payment on selected rows, then application of the old remaining-state moment.
Actual unmarked inputs and plain adapter	The fixed-family plain and inverse input estimates, with actual witness and coefficient dictionaries.
Height-uniform density	Uniform thresholds, bounded windows and the sixth-power lift, using its stated source inputs.
Signed and canonical-source lemmas	Opposite twist signs, complete mask maxima, fixed-modulus nonprincipality, actual restricted coefficient factorization and the padded-radius ledger.
Scale-shortening lemma	The arithmetic dichotomy paying sufficiently shortened plains by the old moment.
Endpoint certificate	A scalar saving below θ on the complete parameter rectangle.

The 48 reported axiom lists contain only `propext`, `Classical.choice`, and `Quot.sound`. This does not constitute an axiom audit of a new complete endpoint theorem. The package manifest, commands, input hashes and receipts identify the exact component scope.

Independent mathematical review of the complete composition is still needed, including its use of the pinned incumbent analytic results and its order of quantifiers. No new complete all-family Lean theorem has been constructed. Clean dependency replay, comparison against the canonical target statements, and independent kernel verification have not been completed. These are validation limitations, separate from the written proof now supplied. Until that review and verification are complete, this manuscript should be described as a proposed analytic proof under review, not as an independently certified improvement.

8 Attribution and source relationship

OpenAI's development supplies the initial 7/8 theorem and the analytic and arithmetic framework. Levy's work supplies the weighted-numerator input adapted by the Cycle25 source. Collet's work supplies the variable moment, detector, quartic optimization, and source architecture used throughout this draft. Our proposed contribution is the local 4.498 refinement, its joint-label analysis, and its proposed endpoint assembly supported by an exact rational calculation. These descriptions do not imply endorsement by any upstream author or maintainer.

This manuscript develops the three candidate notes in the research package [4], the second-revision input and reconstruction work [5], and the new source arguments included in the supplements. It distinguishes their mathematical claims from their current validation status. Research,

code, formalization, and writing received AI assistance; the internal reviews recorded with the package are model-assisted development reviews, not independent human referee reports. The package retains its Apache-2.0 license and upstream notices.

A Reproduction and the certificate format

The package linked in [4] contains three notes under the directory `research/framework/programs/fixed_strip/candidates/`:

- `LOCAL_PHYSICAL_MOMENT_EXTENSION.md`;
- `JOINT_HIGH_ZERO_SQUARED_LABEL_PAYMENT.md`;
- `LOCAL_MOMENT_TRACKER_REFINEMENT.md`.

The first explains the proposed local moment, the second isolates a useful squared-label branch, and the third records the endpoint geometry. In particular, the coefficient 21/5 used illustratively in the second note is not the coefficient claimed in Proposition 2.1, and is not an established stronger moment.

From the first snapshot root `proofs/gao-local-moment-20261011`, with Python and the recorded SymPy version available, run the following commands. These paths belong to that original snapshot, not to the present revision directory.

```
python3 reproduce.py --fetch-pinned-paper
python3 research/tools/local_physical_moment_audit.py --check
python3 research/tools/local_moment_endpoint_audit.py --check
```

The fetched manuscript is a pinned input, checked by its stored digest; it is not a paper authored by this project. The outputs in `research/numerics/results/` record the exact assertions and their scope. Both reports explicitly deny verification of the new full analytic theorem. The present revision also supplies the following additional exact check:

```
python3 endpoint_extra_arithmetic.py --check
```

It verifies 20 rational comparisons used by the written endpoint assembly. The current component replay is documented with `replay_components.py` in the revision README; it has separate compiler and pinned dependency prerequisites.

For completeness, the Bernstein certificate used in Proposition 6.1 is based on the identity

$$f(u,v)=\sum_{i=0}^{m}\sum_{j=0}^{n} b_{ij}\binom{m}{i}u^i(1-u)^{m-i}\binom{n}{j}v^j(1-v)^{n-j}, \qquad 0\le u,v\le 1.$$

The basis functions are nonnegative and their sum is one. Thus $b_{ij}\ge 0$ for all i,j proves $f\ge 0$ on the whole square. An exact affine substitution maps each recorded rational rectangle to that square. Rational reconstruction of the coefficients and exact coverage checks make the conclusion inspectable without trusting floating-point optimization.

Guide to the supplementary source arguments

The following sections are part of the proposed analytic proof. They retain the exact pinned source interfaces so a reader can inspect each dependency. The local and endpoint assemblies are proposed proofs under review. Their distinction from independent certification is stated in Section 7.

B Signed rows, zero extensions, and label reconstruction

The purpose of the reconstruction is to identify one integer row to which the unmarked density theorem applies. It must identify the character on all arguments, including its zeros, and must also explain which arithmetic labels can be recovered from that row. Merely identifying a primitive character on coprime arguments would lose the natural masks.

B.1 The two signs at a common prime

For supported primary elements define $c_n(v) = \mathfrak{s}_v((n))$, the source's zero-extended sextic ideal row. Let $q(\pi, n)$ be the fixed reciprocity phase, so

$$c_\pi(n) = q(\pi, n)c_n(\pi).$$

The phase belongs to the fixed ray dictionary. If $\pi \mid n$, both sides vanish; the identity is not restricted to the unit case. For a common prime with active net exponent $1 \le t \le 5$, it follows that

$$\begin{aligned} c_n(\pi)c_\pi(n)^t &= q(\pi, n)^t c_n(\pi)^{1+t}, \\ c_n(\pi)\overline{c_\pi(n)^t} &= q(\pi, n)^{6-t} c_n(\pi)^{7-t}. \end{aligned}$$

On units this is ordinary multiplication of sixth roots of unity; off units each displayed product is zero. Thus the left coefficient has residue $1 + t$ and the right coefficient residue $1 - t$, modulo six. These are the signs in the actual formal coefficients `leftCoefficient` and `rightCoefficient`. A manuscript convention assigning the conjugate coefficient to the C-side must exchange the side names consistently.

If $t = 5$ on the left, or $t = 1$ on the right, the resulting exponent is six. The source's exact identity is

$$c_n(\pi)^6 = \mathbf{1}_{(n,\pi)=1}.$$

Consequently reducing the exponent to zero moves this factor into the puncture; it does not replace it by one. Multiplying the primewise identities proves the corresponding complete first-row character identity with every canceled prime still punctured.

For one side let $F = \prod_{\pi \mid r} \pi^{e(\pi)}$, choosing the source's primary prime generators, and let $L = \log_Z NF$. The ideal used by the formal source has generator idealGenerator($\mathfrak{r}$), which can differ from the product of chosen generators by a unit. Retain that unit in the effective row, or absorb its character into the fixed six-unit dictionary. No unproved equality of these two generator choices is needed.

B.2 The complete second frequency

For the second common-support ideals D_2, E_2, let G be their common gcd generator. The actual correlation vanishes unless $G \mid j$. After writing $j = Gu$, the nonunit part of the equal-multiplicity

partition forces $V \mid u$, where V is its radical generator. Thus the genuine nonzero frequency is

$$j = GVh', \qquad h' \neq 0.$$

The source's partitioned correlation, divided by its stated normalizer, has modulus at most one, including its extension by zero off the partition. Multiplicativity now combines the first row with this frequency:

$$w = u_0\, eFGVh',$$

where u_0 is the retained unit. Every surviving whole-product character is a fixed ray factor times $c_n(w)$, with the common puncture and all canceled-prime holes. Relative slot characters remain their original fixed factors. Whole-character conjugation or the reflected row $-w$ is handled by whole-character height reflection and the fixed unit list. The dictionary is a finite union of parent translates of the allowed Q_*-character group. The buffered maximum is formed within the actual parent translate; another translate is not added to that maximum. Conjugating only its fixed factor would change this family and is not an allowed replacement.

The exact source ingredients are `SecondCanonicalFrequency.idealCorrelation_zero_outside_common`, `SecondCanonicalNonunit.nonunitFrequencyGenerator_dvd`, and `SecondCanonicalFrequency.actual_ideal_partition_scalar`. The new signed-row lemmas check the primewise identities and their multiplicative combination against these source conventions.

With the actual common main ball of logarithmic radius $K_0 + (d-c)_+ + 6a + \varpi$, the second frequency envelope gives

$$\log_Z Nw \leq M - R - (c-d)_+ + L - 6a - \varpi + \eta.$$

Discarding the favorable $-\varpi$ gives (11). A nonempty integral row window always has $Nw \geq 1$; negative nominal logarithmic ranges are either empty or are filled by the reserved bounded-window allowance. Density is not applied as a count smaller than one on a nonempty window.

B.3 Which labels are divisors of the row?

The first residual projector label s divides both second columns. After the complete second common-support extraction it therefore divides G, and hence divides w. The entire second common radical also divides G. The source proves these assertions as `SecondSourceMask.source_pair_divisor_common` and `divisor_dvd_common`.

The second residual projector label t is different. Its individual terms belong to the artificial separated extension of the genuine correlation, and t need not divide G or w. Freeze it while estimating the row norm. Its signed priority allocation selects plain divisors and freezes prime slots. If a selected plain is $n = dv$, then

$$\psi_w(dv) = \psi_w(d)\psi_w(v).$$

The factor $\psi_w(d)$ is retained outside, even if it is zero. The quotient v may still be divisible by a prime of d; no new coprimality condition is introduced. This explains why the primitive row and its buffered bin survive allocation. Its divisor-dyad count is paid together with the two exact normalization scalars in B3.67–B3.69.

Among the first common primes, those active with nonzero row residue occur in F, and the selected mask-divisor primes occur in e. The remaining first radical has logarithmic width at most

$$p - R - E + R_{\mathrm{zero}}.$$

This accounts separately for inactive net-zero primes and for active primes whose row phase cancels. They cannot be called divisors of w. Once their radical is fixed, all remaining first labels are supported on it together with $\operatorname{rad}(w)$.

For completeness, a uniform support-counting bound follows from Rankin's inequality. If $NQ \le Z^B$ and $NI \le Z^T$, then for any $s > 0$,

$$\#\{I : \operatorname{rad} I \mid Q,\ NI \le Z^T\} \le Z^{sT} \prod_{\mathfrak{p} \mid Q} (1 - (N\mathfrak{p})^{-s})^{-1}.$$

Choose sT small. Each sufficiently large prime factor in the product is at most $(N\mathfrak{p})^b$ for any prescribed $b > 0$; the finitely many remaining primes contribute a constant. Choosing bB small proves a bound $O(Z^\epsilon)$ uniformly in the support ideal. The field has only $O(X)$ ideals of norm at most X, so counting the missing radical costs its displayed norm exponent. Fixed finitely many slot assignments, bounded-range powers and priority allocations have the same arbitrarily small power allowance after the slot system is fixed.

This proves the general label reconstruction used in the high norm. It retains the first labels until the joint row count, and it does not charge the second projector count twice. Nonprincipality and primitive conductor bounds are supplied by the following fixed-dictionary section; they are not consequences of the size bound on w alone.

C The fixed dictionary behind the actual signed rows

This section concerns the local family whose parent character and ray modulus are fixed before the scale. It does not assert a uniform density estimate for arbitrary parent characters with growing conductor. That stronger assertion is unnecessary here: the moving factors in both transformations are retained in the complete integer row.

Write $m_0 = 2(1 - \omega)$. Let Q be the *literal* fixed ideal used by the source's fixed-inducing-row predicate. Its hypotheses include $Q \subseteq (72)$ and the stated containments in the original fixed ray moduli. The relative slot characters have presentations of modulus dividing Q. These containments use the ideal convention that $Q \subseteq \mathfrak{m}$ means $\mathfrak{m} \mid Q$. For the actual initial family F, this ideal is

$$Q = \operatorname{internalQ}(F.\text{modulus} \cap (72))\ \eta_0 = F.\text{modulus} \cap (72) \cap \mathfrak{q}_{\eta_0},$$

as defined by `NaturalFixedRaySourceFixedIdeal.sourceFixedIdeal` and `sourceExceptional`. Thus the needed containments belong to the literal source family; they are not a later enlargement of its exceptional set. In a transformed child this ideal remains fixed even though the separate whole-product character can move.

A finite collection chosen before arithmetic labels. Take the primary ray characters modulo 12, the six unit-row characters modulo 36, and the allowed relative slot characters, together with their inverses. On ideals prime to Q their products belong to the finite group of ray characters modulo Q. Multiply these products by each of the original fixed parent characters; the resulting finite collection is a union of fixed translates of that group. The parent itself need not have modulus dividing Q. Different zero extensions at primes dividing Q introduce only finitely many additional presentations. Thus there is a finite list $\{\eta_\lambda : \lambda \in \Lambda\}$ containing every product that will occur, chosen before the common supports, frequencies, or rows. Products over an arbitrarily large number of active primes do not enlarge this list: the reciprocity phases are characters of a single

fixed residue group. They do not introduce one new independent character for each prime. Keep distinct translates of the Q-character group separate when defining zero bins. The finite union is used to make constants simultaneous; it does not imply that eligibility for one translate makes all other translates nonprincipal. Partitioning the fixed outer labels by their parent translate costs a fixed finite factor in the eventual joint count.

The source realizes the primary phases by `FixedRay.primaryRayCharacter`, whose modulus is (12), and unit phases by `HeckeUnitRows.character`, whose modulus is (36). For the relative factors, the exact source formula is

$$\nu_\theta = \chi_\theta \eta_0^{-1}, \qquad Q \subseteq M,\ Q \subseteq \mathfrak{q}_{\eta_0} \implies Q \subseteq \mathfrak{q}_{\nu_\theta}.$$

This is `NaturalFixedRaySourceCharacters.relativeCharacter_fixedQ`. The allocation theorem `AllocatedRayDictionary.originalImage_fixedQ` preserves it for every surviving slot. The full source identities `both_common_rectangles` retain the original relative characters, profiles and independent heights; allocation changes which original slots remain, not their character dictionary.

Exact coefficient identity. Use the signs and primary-generator conventions of the signed-row reconstruction. For either first-side orientation the complete second integer row is

$$w = u\, eF_{\pm}GVh \neq 0,$$

where the fixed unit u may instead be absorbed into the finite list. Every actual residual character coefficient, including a surviving relative slot twist, is of the form

$$c_{\eta_\lambda}(I)\, \mathfrak{s}_{m_0^6 w}(I) \mathbf{1}_{(I,\mathfrak{R})=1}. \tag{FD}$$

Here $\mathfrak{R}$ is the retained natural and common puncture. This is an identity on all ideals, including the zero values. On supported ideals it follows by evaluating the signed identities at the primary generator; on unsupported ideals both sides vanish because of m_0. A canceled exponent at an active prime produces a factor $\mathbf{1}_{(I,\mathfrak{p})=1}$ and therefore belongs to $\mathfrak{R}$. The second extraction contributes GVh multiplicatively to the same integer row. The complete second coprimality projector is expanded with its original signs. Extracting one of its assigned divisors d uses $\psi(dn) = \psi(d)\psi(n)$; it does not impose an additional condition $(d, n) = 1$. A zero outside coefficient annihilates that term. These facts are exactly the distinctions made in the signed-row reconstruction.

The two first-side orientations can be treated separately, at a fixed factor of two. Within the square of one first-side column the reflected second rectangle has the row $-w$, which is covered by the same finite unit-character list. There is no claim that F_+ and F_- are equal. If a whole rectangle is conjugated, retain that conjugation of the whole character: its zero maximum is unchanged after reflection of the height coordinate. Do not replace $\overline{\eta\, \mathfrak{s}_w}$ by $\overline{\eta}\, \mathfrak{s}_w$. Such a replacement could change the parent translate and its exceptional rows.

Choose the natural character $\chi_{\lambda,w}$ for the unmasked part of (FD). The natural constructor gives

$$N\mathfrak{q}_{\chi_{\lambda,w}} \leq N\mathfrak{q}_{\eta_\lambda}\, N(m_0) N(72)\, Nw \leq KNw, \tag{FC}$$

with one fixed K. Its ideal coefficients equal those of `momentData` η_λ; the new theorem `natural_row_moment_dictionary` verifies this equality. One must use this natural representative for (FC), since the abstract choice inside `RowData.character` has no stated conductor bound. The primitive inducing character has still smaller conductor.

Literal exclusion survives the change of presentation. For a nonzero mask and row, the source proves

$$\mathrm{FI}_Q(\eta; m_0 r, 1, w) \iff \mathrm{FI}_Q(\eta; m_0, 1, w), \tag{FE1}$$

by `FixedRowMask.fixedInducingRow_mul_mask_iff`. If $Q \subseteq \mathfrak{q}_\nu$, it also proves

$$\mathrm{FI}_Q(\eta\nu; m_0, 1, w) \iff \mathrm{FI}_Q(\eta; m_0, 1, w), \tag{FE2}$$

by `FixedRayInducingTransport.fixedInducingRow_product_iff`. The reverse implication is not obtained by canceling a possibly zero character value. Multiplication by ν^{-1} recovers the original character with the additional modulus mask of ν; (FE1) then removes that mask from the *inducing predicate*. This is why the equivalence remains valid on the literal zero-extended family.

Apply (FE1), (FE2), and (FD) to the actual source exclusion. They identify it with the same predicate on $\chi_{\lambda,w}$. In particular every allowed relative twist in this same parent translate is nonprincipal on an eligible row. No conclusion is made here about another parent translate. The source states this consequence directly as `NaturalFixedRaySourceCharacters.sector_nonprincipal`. For the second common extraction, `SecondExceptionalFamily.Family.inducing_iff` and `paired_inducing_iff` implement precisely the reduced-row and reflected-row versions of this transport. The ideal Q is not enlarged. It would be incorrect simply to replace the source exclusion by nonprincipality of one preferred primitive representative.

Zero bins and the density input. For one fixed parent translate, take the maximum of the zero real parts over its allowed Q-character twists, and use the buffered height boxes of the source. Whole-character inverses have the same maximum and need not be modeled by an additional translate. Each actual character in that maximum is a masked presentation of one of (FD). Finite Euler factors are nonzero in $\Re s > 0$, so its zeros in these boxes equal those of its primitive inducing character. Removing sixth powers, $w = ub^6$, introduces further Euler holes and does not change these positive-real zeros. The fixed-dictionary density proposition therefore applies to the selected free core u, followed by the height-uniform sixth-power lift.

If a bin is selected by the maximum over all allowed labels in this translate, some label attains that maximum. The union over this *fixed* finite list costs only a constant. The upper side of the buffered bin bounds every surviving prime-slot character, while the lower side supplies the zero used for density. This step neither sums over the moving arithmetic labels nor freezes them before counting. Those labels are recovered from w and the missing radical by the signed-row reconstruction.

Consequently, for $7/10 \le \delta \le 3/4$, the actual eligible integer rows in the selected high bin satisfy

$$\#\{w : Nw \le H\} \ll_\varepsilon (1+T)^J H^{\rho(\delta)+\varepsilon}, \qquad \rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)},$$

up to the stated arbitrarily small bin-buffer loss. The constants and threshold precede T, every live common-support label, and H. Polynomial-size masks do not enter the dictionary size or the primitive conductor in (FC); their smooth estimates are paid separately by uniform Euler deletion. The pointwise upper bounds use the common buffered bin, whereas the count only needs the label attaining its lower edge.

Fourier kernels whose parameter varies with labels. The common Fourier argument also permits a scale parameter A_ℓ depending on a retained label ℓ. Uniform Euler derivative bounds

give Fourier densities satisfying $|f_A(t)| \le C(1+|t|)^{-2r}$, uniformly in A. Choose once and for all $d\mu(t) = C(1+|t|)^{-2r}\,dt$ and write $f_{A_\ell}(t)\,dt = a_\ell(t)d\mu(t)$, where $|a_\ell(t)| \le 1$. The integral of the joint vector then contains the diagonal contraction $\mathrm{diag}(a_\ell(t))$. Its operator norm is at most one, so Minkowski's inequality gives exactly the same joint bound, with no extra label count. This avoids the unjustified assertion that the individual Fourier measures for distinct A_ℓ are equal.

This establishes the fixed-dictionary, conductor, mask, and literal eligibility bridge from the actual signed-row identities. The remaining assembly of the physical moment must still verify its normalization, comparison and exceptional terms, and its use of the reserved exponent margins. Those conclusions do not follow from this bridge alone.

D The unmarked plain moment and density for a fixed character dictionary

This section closes the zero-slot plain-moment input for a fixed finite character dictionary. It does not identify the varying rows in the proposed physical moment with such a dictionary; that separate identification must preserve every Euler hole and the stated constants.

Let $\mathcal{O} = \mathbb{Z}[\omega]$ be the Eisenstein integers and put $m_0 = 2(1-\omega)$. A free row means a nonzero element u whose principal ideal has every prime valuation less than six. Fix a nonempty finite set Λ (the empty case is immediate) and characters η_λ, $\lambda \in \Lambda$. For each u choose the *natural* row character $\chi_{u,\lambda}$, so that

$$c_{\chi_{u,\lambda}}(I) = c_{\eta_\lambda}(I)\,\mathfrak{s}_{m_0^6 u}(I), \qquad \mathrm{N}\mathfrak{q}_{u,\lambda} \le K_\lambda\,\mathrm{N}(u), \quad K_\lambda = \mathrm{N}\mathfrak{q}_{\eta_\lambda}\mathrm{N}(m_0)\mathrm{N}(72).$$

Here $\mathfrak{s}$ is the literal extended sextic ideal character, including its zero values at ramified primes. The conductor estimate refers to the natural representative obtained from the conductor-bounded row constructor. The abstract `RowData.character` is chosen by a different existence theorem and does not carry this bound. Its coefficients agree with the natural representative, which is all the moment estimates require. Whole-character conjugation preserves the modulus and reflects the zero height. It can be handled through this symmetry; it must not be confused with conjugating only the fixed factor η_λ.

D.1 The zero-slot plain input

For $n \le 2$, $\sigma \in [0,1]$, $t \in \mathbb{R}$, write

$$P_n(\chi; D, \sigma, t) = D^{-1/2} \sum_{I \ne 0} c_\chi(I)(\log(\mathrm{N}I/D))^n W(\mathrm{N}I/D)(\mathrm{N}I/D)^{-\sigma+it},$$

where W is the fixed positive annular detector profile, supported in $[1/4, 9/4]$. Its logarithmic derivatives occurring here are those of the existing detector construction.

Plain-moment proposition. Assume the contradiction branch $\beta \ge 51/100$. For each $\epsilon > 0$ there are $C > 0$, $J \in \mathbb{N}$, and U_0, depending on the fixed dictionary and ϵ, such that for every $U \ge U_0$, every height $H \ge 0$, every unmarked detector fiber whose right length satisfies $m \le 3/4$, and whose finite row set $\mathcal{R}$ satisfies $\mathrm{N}(u) \le U$ and the displayed coefficient dictionary, one has

$$\sum_{u \in \mathcal{R}} |P_j(\chi_{u,\lambda}; U^m, \sigma, t) P_k(\chi_{u,\lambda}; U^m, \sigma, t)|^2 \le C(1+H)^J U^{\max(1,2m)+\epsilon}, \tag{P}$$

for $j + k \le 2$, $\sigma \in [0,1]$ and $|t| \le H$. The threshold precedes the height quantifier. No lower bound on $\mathrm{N}(u)$ and no prime-slot supply is assumed. In the formal theorem $\mathcal{R}$ is the row set of an unmarked detector fiber; the proof of its plain estimate only uses its row bound, coefficient dictionary, and m.

Proof. Fix a label and choose, for example, $Q = \mathfrak{q}_\eta(72)$. Then $Q \neq 0$, $Q \neq \mathcal{O}$, $Q \subseteq (72)$, and $Q \subseteq \mathfrak{q}_\eta$. These are the exact fixed-modulus gates of the incumbent analytic induction. For $d = \min(\epsilon/4, 1/4) > 0$, its terminal certificate with $\kappa = \max(3/4, 2\beta - 1)$ supplies the zero-slot bound at error d. The zero slot profile is a legitimate input to that certificate; no improvement of the physical moment is used. The fixed character modulus satisfies $\mathrm{N}\mathfrak{q}_\eta \le U^d$ for all sufficiently large U. The natural initial state has width

$$1 + d + \max(0, 2m - 1) = \max(1, 2m) + d \le 2.$$

Consequently the incumbent zero-slot estimate, followed by the uniform Schwartz-profile bounds, proves (P) on rows satisfying `initialKeep`, with exponent $\max(1, 2m) + 2d$. The profile control contributes $C(1 + H)^J$ and does not change the scale threshold. The legacy proof interface is reached by adding one auxiliary slot of width $7/37$ with identically zero profile and always selecting the empty product. Every legacy structure field, including its numerical supply field, is then proved; no prime sum or supply property is imposed on the original family.

The omitted free rows form a fixed finite set E_η. Indeed, the fixed-inducing-row theorem puts all their prime factors in the finite support of Q, and free-row valuations are at most five. There are only finitely many resulting ideals and six possible units per principal ideal. This is a fixed finite set, not the whole interval of small rows.

The exceptional contribution has the required size without cancellation. The detector Schwartz profiles have a uniform absolute bound A for all $n \le 2$, $\sigma \in [0,1]$, and all frequencies. Since $\#\{I : 0 < \mathrm{N}I \le x\} \le 128x$, coefficient norms are at most one, and the profile vanishes above $9/4$, the normalized finite sum gives

$$|P_n(\chi; D, \sigma, t)| \le 288A\sqrt{D}$$

for every character, including principal characters. Therefore

$$\sum_{u \in \mathcal{R} \cap E_\eta} |P_j P_k|^2 \le |E_\eta|(288A)^4 D^2 \le |E_\eta|(288A)^4 U^{\max(1,2m)+\epsilon}.$$

This proves (P) on all rows. Taking maxima of finitely many thresholds and degrees, and a sum of the finitely many constants, proves the simultaneous statement for Λ. The compiled theorem `fixed_unmarked_plain` proves this argument for one label; finite label closure is an ordinary finite maximum argument.

D.2 From actual witnesses to a height-uniform density bound

Let $N(Y, a, T)$ count free rows u, $\mathrm{N}(u) \le Y$, for which at least one dictionary character has a zero with real part at least a and imaginary part of absolute value at most T. In this paragraph $17/20 \le a \le 7/8$. Use the incumbent bound $\beta \le 7/8$. If $\beta < a$, the nonexceptional contribution is zero; the fixed exceptional set is handled by its constant cardinality. The lower hypothesis $\beta \ge 51/100$ in (P) is automatic whenever the nonexceptional contribution is nonempty and hence is not an extra theorem assumption on the desired zero-free result.

Density proposition. For every $\varepsilon > 0$ there are C, J, Y_0, fixed before a and T, such that for all $Y \geq Y_0$ and $T \geq 0$,

$$N(Y, a, T) \leq C(1+T)^J Y^{\rho(2a-1)+\varepsilon}, \qquad \rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)}. \tag{D}$$

The constant depends on the fixed dictionary. The same assertion for a scale-dependent dictionary does not follow from this proposition.

Proof. Set $K = \max(1, \max_\lambda K_\lambda)$ and $U = KY$. Then both the row norm and every natural character modulus are at most U. This is a fixed multiplicative padding; no power of Y is lost. Remove the finite union $E = \bigcup_\lambda E_{\eta_\lambda}$. On the remaining rows every natural character is nonprincipal, by the incumbent nonprincipal gate. The finite union contributes $O(1)$ to (D), so there is no need to construct detector witnesses for its members.

Choose small detector and moment errors before the scale parameter. The actual inverse moment for each fixed `momentData` family and (P) now give both moment inputs with one bound $C(1+H)^{J_0}$. Both the constant and degree are independent of the detector zero bin, the dyadic indices, and U. The actual detector lengths satisfy $m \leq 1/2 + 75\epsilon_{\mathrm{det}} \leq 23/40$, so (P) applies. Choose $\tau_{\mathrm{det}} > 0$ *after* J_0, sufficiently small for the supported witness construction and for $\tau_{\mathrm{det}}(J_0+1)$ to fit within the allotted final error.

First suppose $T \leq U^{\tau_{\mathrm{det}}}$. For each row, apply the existing finite buffered-bin lemma to its finite character family with height step $U^{\tau_{\mathrm{det}}} > 2$ and mesh $e > 0$. It gives an integer $1 \leq i \leq I$, $I = \lceil 49/(100e) \rceil$, and a real grid value $b = 51/100 + ek$ such that

$$b \leq M(3iU^{\tau_{\mathrm{det}}}) < b + e, \qquad M(3(i+1)U^{\tau_{\mathrm{det}}}) < b + 2e.$$

Here M is the actual maximum of zero real parts in the specified box, with the usual $51/100$ fallback. Because the original row has a zero with real part at least a at height at most T, monotonicity gives $b > a - e$. The incumbent upper bound gives $b \leq 7/8$. There are only $O_e(1)$ possibilities for (i, k), so partition the rows by these values. For each part set $\delta_b = 2b - 1$ and fix its balanced cutoff $t_* = 5/(5 - 2\delta_b)$ before constructing witnesses.

The actual supported-witness theorem, with base scale $Z = U$ and $d = 1$, produces witnesses in each nonempty part. Its hypotheses are now literal: nonprincipality, modulus at most U, the two buffered maxima, and the natural-row coefficient identity. It uses frequency allowance $U^{\tau_{\mathrm{det}}/2}$, and the witness frequency is bounded by

$$H_i = 2\pi U^{\tau_{\mathrm{det}}/2} + 3iU^{\tau_{\mathrm{det}}} \ll_e U^{\tau_{\mathrm{det}}}.$$

Choose the auxiliary small parameters in the stated detector budget $12e(22+2) + 8\kappa_{\mathrm{aux}} + 2\eta_{\mathrm{aux}} \leq \epsilon_{\mathrm{det}}/2$ and $4\tau_{\mathrm{det}} < \eta_{\mathrm{aux}}$. They can all be chosen positively, with $e < 1/1000$, and then τ_{det} decreased further as necessary. No height variable occurs in the resulting eventual threshold.

Partition once more by the witness label and its two dyadic indices, all constructed using this fixed t_*. The no-slot inverse and plain estimates yield the already compiled balanced count

$$|\mathcal{R}_{\mathrm{fiber}}| \ll (1+H_i)^{J_0+1} U^{\rho(\delta_b)+156\epsilon_{\mathrm{det}}+\epsilon_m}.$$

The fixed label count and the two dyadic counts cost an arbitrary positive power of U. Since $b > a - e$ and ρ is decreasing, differentiation on the relevant compact interval gives $\rho(2b-1) \leq \rho(2a-1) + 3e$. Choose $\epsilon_{\mathrm{det}}, \epsilon_m, e$ and the dyadic padding so their sum with $\tau_{\mathrm{det}}(J_0+1)$ is less than ε. Summing

the finite partitions proves $N(Y,a,T) \ll U^{\rho(2a-1)+\varepsilon}$ in this case. The replacement $U = KY$ only changes the fixed constant.

If $T > U^{\tau_{\mathrm{det}}}$, the elementary lattice count is enough: $N(Y,a,T) \ll U$. Since $\rho(2a-1) > 0$ and $U \geq 1$, for any integer $J \geq 1/\tau_{\mathrm{det}}$,

$$U \leq T^J \leq (1+T)^J U^{\rho(2a-1)+\varepsilon}.$$

This proves (D) for arbitrary T, with a common threshold fixed before that height. Bounded Y are covered by enlarging the constant using the finite row ball, uniformly in T and a.

When this result is used at an external physical height $T = Z^{\tau_{\mathrm{out}}}$, choose τ_{out} only after the final degree J in (D) has been fixed, so that $J\tau_{\mathrm{out}}$ fits the physical error allowance. The external exponent τ_{out} is distinct from the internal detector exponent τ_{det} chosen in this proof. Reusing a single exponent for both purposes would make the parameter selection circular.

The exponent satisfies $\rho(\delta) \geq 2/7 > 1/6$ for $\delta \leq 3/4$. Thus the weighted sixth-power raw-row lifting proved in `HeightUniformDensity.lean` applies to (D). This completes the analytic unmarked density input for a fixed dictionary. Applying it to the joint physical row still requires the exact signed coefficient identity, a fixed-dictionary reduction for every varying label, and the stated conductor and mask accounting.

D.3 Formal verification scope

The new module `PlainUnmarkedAdapter.lean` contains nine audited objects: the natural-row coefficient dictionary, the legacy embedding, the retained estimate, the actual zero-slot input, the retained eventual field, finiteness of exceptional free rows, the absolute detector-polynomial bound, the finite-row absolute bound, and the complete eventual plain field. All nine printed axiom sets contain only `propext`, `Classical.choice`, and `Quot.sound`. Compilation reuses development dependency objects; this is not a clean source replay or an independent checker certificate. The assembly of (D) above is a written mathematical proof, not an additional Lean theorem in this module.

E Uniform deletion and common Fourier transport

The following estimates resolve two bookkeeping questions in the proposed argument. They neither identify a transformed character nor supply a missing mean-value estimate. The field throughout is the fixed Eisenstein field. Constants may depend on the fixed field, the indicated compact supports, and the requested error exponent.

Proposition E.1 (Uniform finite Euler deletion)**.** *Fix $B \geq 0$, $\sigma_0 > 0$, an integer $j \geq 0$, and $\epsilon > 0$. Let $Z \geq 2$, let $\mathfrak{q}$ be a squarefree integral ideal with $N\mathfrak{q} \leq Z^B$, and choose complex numbers $|a_{\mathfrak{p}}| \leq 1$. Put*

$$E_{\mathfrak{q}}(s) = \prod_{\mathfrak{p}\mid\mathfrak{q}} (1 - a_{\mathfrak{p}}(N\mathfrak{p})^{-s}).$$

Then $E_{\mathfrak{q}}$ is nonzero on $\Re s > 0$, and, uniformly in $\mathfrak{q}$, the coefficients, and $\Im s$,

$$|E_{\mathfrak{q}}^{(j)}(s)| + |(E_{\mathfrak{q}}^{-1})^{(j)}(s)| \ll_{B,\sigma_0,j,\epsilon} Z^{\epsilon} \qquad (\Re s \geq \sigma_0).$$

Proof. Every factor is nonzero because $|a_{\mathfrak{p}}(N\mathfrak{p})^{-s}| < 1$ in the positive half-plane. For $\Re s \geq \sigma_0/2$, both the product and its reciprocal are bounded by

$$\prod_{\mathfrak{p}|\mathfrak{q}} (1 - (N\mathfrak{p})^{-\sigma_0/2})^{-1}.$$

Given $b > 0$, the factor at every sufficiently large prime norm is at most $(N\mathfrak{p})^b$: its logarithm is $O((N\mathfrak{p})^{-\sigma_0/2}) = o(\log N\mathfrak{p})$. The finitely many smaller primes contribute a constant, independently of the chosen ideal and coefficients. The product is consequently $O((N\mathfrak{q})^b)$. If $B > 0$, choose $bB \leq \epsilon$; if $B = 0$ the product is empty. Cauchy's derivative estimate on disks of radius $\sigma_0/2$ proves the claim for every fixed j. □

Thus, once the exact identity $L_{\text{masked}}(s) = L_{\text{primitive}}(s)E_{\mathfrak{q}}(s)$ is established, extra polynomial-size holes preserve its positive-real zeros and cost an arbitrarily small power in a contour argument staying in a fixed positive half-plane. For a nonprincipal character the identity also applies at $s = 1$. A principal pole must still be treated separately. This argument places no small-conductor hypothesis on the *masked presentation*. It applies the analytic estimate to the primitive presentation and restores the finite factors afterward. For the raw row family one still has to prove the primitive-conductor bound and the exact coefficient identity; these are not consequences of a product estimate.

For a prime polynomial with relative support $[c, C] \subset (0, \infty)$, bounded coefficient modulus, and central normalization $P^{-1/2}$, deleting primes dividing $\mathfrak{q}$ changes it by

$$O(P^{-1/2} \log(2N\mathfrak{q})) \qquad (P \geq 1).$$

Indeed there are at most $\log(N\mathfrak{q})/\log 2$ such primes. Fixed powers of $\log N\mathfrak{p}$ add only powers of $\log Z$ on polynomial-size scales. Real powers on the normalized support have bounded modulus, and imaginary powers have modulus one. This error is $O_\epsilon(Z^\epsilon)$ and is compatible with a squared pointwise bound $P^\delta Z^\epsilon$ whenever $\delta \geq 0$, after dividing the error allowance between the original estimate and this correction. No deletion of an arithmetic mask by positivity is involved.

Proposition E.2 (Fourier transport with labels inside the norm)**.** *Fix a compact box $K \subset (0, \infty)^d$ and $J \geq 0$. Suppose that a family of smooth profiles H_A is supported on one larger compact box and has uniformly bounded Euler derivatives through order $2r$, where $2r > J + d$. Then it has representations*

$$H_A(y) = \int_{\mathbb{R}^d} \prod_{k=1}^{d} y_k^{it_k} \, d\mu_A(t), \qquad \int_{\mathbb{R}^d} (1 + |t|)^J \, d|\mu_A|(t) \leq C_J,$$

uniformly in A. The measure is chosen before any arithmetic label or row is evaluated.

Let Λ be a finite set of retained arithmetic labels, and let $F_t = (F_t(\lambda, h))$ be a measurable family in $\mathcal{H} = \bigoplus_{\lambda \in \Lambda} \ell^2(R_\lambda)$. If

$$\|F_t\|_{\mathcal{H}} \leq B(1 + |t|)^J,$$

then

$$\left\| \int F_t \, d\mu_A(t) \right\|_{\mathcal{H}} \leq C_J B.$$

In particular there is no factor $|\Lambda|$ in this transport. The same conclusion holds for weighted direct sums with fixed nonnegative arithmetic weights.

Proof. Pass to logarithmic coordinates $x_k = \log y_k$ and apply Fourier inversion. Integrating by parts with $(1-\Delta)^r$ bounds the Fourier density by $C(1+|t|)^{-2r}$. Compact support bounds the necessary L^1 derivative norms uniformly. The asserted weighted integrability follows because $2r > J + d$.

The second assertion is the triangle inequality for the Bochner integral in the Hilbert space $\mathcal{H}$:

$$\left\| \int F_t \, d\mu_A(t) \right\|_{\mathcal{H}} \le \int \|F_t\|_{\mathcal{H}} \, d|\mu_A|(t) \le B \int (1+|t|)^J \, d|\mu_A|(t).$$

The assumption supplies integrability. Alternatively, all row sums may first be truncated, with the same estimate, followed by a limit. □

The required uniform derivative assertion holds for a kernel $K(A\prod_k y_k^{e_k})$ if its Euler derivatives are bounded on $(0,\infty)$ and the exponents e_k are fixed. Explicitly,

$$D^\beta K\left(A\prod_k y_k^{e_k}\right) = \left(\prod_k e_k^{\beta_k}\right)(x\partial_x)^{|\beta|}K(x)\Big|_{x=A\prod_k y_k^{e_k}}.$$

Cutoffs on the fixed box and normalized norm powers introduce only fixed seminorms and polynomials in their independent heights. Relations among arithmetic norms restrict evaluation points inside the box; they do not redefine μ_A for each label. Discontinuous coprimality, valuation, or zero-bin indicators remain in the arithmetic vector F_t and are never differentiated.

The substantive premise is a bound on the *joint* vector F_t. Separate per-label bounds cannot be substituted for it without their label count. Conversely, once an exact reconstruction bounds the joint sum over (λ, h), the Fourier step does not charge that label count again. The preceding statement justifies this last passage but does not prove the reconstruction or the required pointwise bounds.

There is also a useful version when an outer arithmetic label changes the parameter A. The integration-by-parts proof gives a single pointwise majorant for the Fourier densities, $|f_A(t)| \le C(1+|t|)^{-2r}$. Choose the positive measure $d\mu(t) = C(1+|t|)^{-2r}\,dt$ before the labels. Write $f_{A(\lambda)}(t)\,dt = a_\lambda(t)\,d\mu(t)$, where $|a_\lambda(t)| \le 1$. For each t, multiplication by $a_\lambda(t)$ is a diagonal contraction on the joint Hilbert space. The same bound therefore follows with this one common dominating measure; equality of the label-dependent Fourier densities is not needed. This is used only for labels fixed outside the sum being Poisson-transformed. It does not permit a discontinuous row-dependent factor to be inserted into that sum before Poisson summation.

Quantifier order for height losses. If a density input is eventually bounded by $C(1+T)^J H^\rho$, its threshold must be independent of T before one substitutes $T = Z^\tau$. An assertion proved only for each fixed T with an unspecified threshold $H_0(T)$ is insufficient. With a uniform threshold, bounded H are filled using the lattice count and $(1+T)^J \ge 1$, without a new height dependence. Fix the zero buffer, the finitely many profiles, and their analytic height exponents first; then choose τ so that the total τJ lies below the reserved power saving. Frequencies outside the retained range require the common Fourier tail estimate. The pointwise and density results must be checked to have precisely this uniform order; it is not supplied by the Fourier identity alone.

F Restricted canonical norms and the exact-width correction

This section explains why the low-row argument applies to the actual allocated character sums. The key point is that the low indicator is a restriction of a positive row norm. It never becomes a smooth profile, a coefficient to be differentiated, or an input to Poisson summation.

F.1 The actual coefficient and radial factorization

Freeze a separated Fourier frequency and every arithmetic label outside the current row sum, including the second projector label. After its complete signed allocation, one positive rectangle norm is

$$\sum_z \mathbf{1}_{\mathrm{keep}(z)}\mathbf{1}_{\mathrm{low}(w(z))}\rho(z)\left|P_1(z)P_2(z)\prod_{i\in J}Q_i(z)\right|^2, \qquad \rho(z)\ge 0.$$

The fixed dictionary includes every relative sector in the current parent translate. It gives a common bin for the slots in this expression. The character is the exact one from Section B.

Let r be the source's canonical radial state. Its keep predicate is $z\neq 0$ together with the literal exclusion for the moving character τ and the fixed ideal Q_*. Define r_{low} by replacing only that predicate with $\mathrm{keep}(z)\wedge\mathrm{low}(w(z))$. Its smooth radial profile, scale, whole-product character and punctures remain the same. Its nonzero-row and nonexceptional conditions follow immediately from those of r, so it is an admissible restricted natural state of the same width.

The source identities apply to precisely this object:

1. `AllocatedNaturalRadial.common_allocated_normalized` expresses the allocated row as the product of two natural plain sums and the remaining natural prime slots. The two scale phases have modulus one; common holes remain as the excluded-prime character.

2. `allocated_radial_bound` and its shared mask estimate allow an arbitrary row keep predicate. Thus mask expansion can use r_{low} and preserves the low restriction in every daughter. Its plain scales are divided by their assigned ideals, and forced prime slots are frozen.

3. `AllocatedChildren.commonChild_energy` identifies the source's actual natural product with its allocated energy for any such radial state. The new `actual_common_child_restricted` lemma checks this identity on the canonical low rows.

4. `AllocatedRayDictionary.common_clipped_energy` reindexes surviving slots to their original indices. It retains their relative characters and replaces the frequency by the common shift plus the original individual height. Its keep argument is arbitrary and therefore still contains the low predicate.

These are coefficient equalities before the moment estimate; arbitrary absolute-value coefficients have not replaced the canonical slots.

For any nonnegative row weight, Cauchy applies with the same restriction on both factors. Alternatively, one can first form a positive norm and then split it into low and high parts. Neither operation asks for a regularity property of the indicator. The smooth Fourier measure remains outside the restricted arithmetic sums, as in Section E.

F.2 Payment before removing the restriction

Choose one greedy subset $R\subseteq J$ from the actual clipped lengths. On the low rows the buffered estimate gives, for each removed slot,

$$|Q_i(z)|^2\le B_i Z^{k_0 z_i}, \qquad k_0=7/10.$$

The factors B_i contain the allocated small losses and height allowance. Multiplying these pointwise inequalities gives

$$\sum_z \mathbf{1}_{\mathrm{keep}(z)}\mathbf{1}_{\mathrm{low}(w(z))}\rho(z)|P_1(z)P_2(z)\prod_{i\in J}Q_i(z)|^2$$
$$\le \Big(\prod_{i\in R}B_i\Big)Z^{k_0\sum_{i\in R}z_i}\sum_z \mathbf{1}_{\mathrm{keep}(z)}\mathbf{1}_{\mathrm{low}(w(z))}\rho(z)|P_1(z)P_2(z)\prod_{i\in J\setminus R}Q_i(z)|^2.$$

Drop the low indicator only in the last nonnegative sum. It is now the full canonical remaining-state norm of the old moment, with the same moving character and fixed ideal. This is the exact factorization proved by `restricted_energy_remove` and used by `restricted_paid_incumbent`.

The two rates serve different roles. Removing one unit of slot width reduces the old capacity defect by $9/2$, but its low-bin squared cost is $7/10$. Greedy removal therefore costs at most $(7/45)F_{\mathrm{act}} + (7/10)\eta_{\mathrm{mesh}}$. If the slots run out first, the existing `ZeroAt` estimate applies without a plain-capacity condition. There is no extra payment for a residual plain defect. Bounded prime-slot lengths are handled directly; the buffered theorem's large-length hypothesis is not applied at length one. For a fixed finite slot list, per-slot losses are chosen from a single prescribed aggregate budget.

F.3 Actual width versus nominal width

The canonical state uses the exact width

$$M_{\mathrm{act}} = \log_Z(\text{dyadic row scale}) + \log_Z N\mathfrak{q}_\tau.$$

It can be smaller than the nominal ledger width M_{nom}. Put $d = M_{\mathrm{act}} - M_{\mathrm{nom}} \le \eta_2$. The source's deleted-child capacity estimate gives $F_{\mathrm{act}} \le (F_{\mathrm{nom}} - d)_+$, so it is the combined quantity $d + rF_{\mathrm{act}}$ which must be bounded. The elementary inequality (7), checked in `width_plus_low_defect`, bounds it by $\eta_2 + rF_{\mathrm{nom}}$. Use $r = 7/45$ on low main children and $r = 1/6$ on the extracted errors.

For the padded actual main radius (12), direct substitution gives

$$M_{\mathrm{nom}} = M - (c-d)_+ - 6a - \varpi - g_2 + t_2$$

and

$$(A - c - c_2 - M_{\mathrm{nom}}) - (A - M) = 6a + \varpi - \min(c,d) - (c_2 - g_2) - t_2 \le 6a + \varpi.$$

These identities also hold with a fixed parent modulus width q, writing $M = m + q$ and putting m in the first nominal frequency formula. They are checked in `RestrictedCanonicalSource.lean`. The nominal low defect is consequently at most $g + 6a + \varpi + 2\theta_N$ plus the fixed presentation padding. The source's inner allowance is still $M_{\mathrm{nom}} + \Delta$, with $\Delta = b_2 - p_2 + B_c + a \ge a$. The scale cancels in that subtraction. Thus the strict reserve (8) survives the exact-width correction, with the padding chosen from it.

Finally, the proposed radius is already the formal common radius

$$\texttt{mainCommonRadius} = Z^{K_0 + (d-c)_+ + 2\sigma + \delta + \mathrm{reserve}}.$$

Take $\sigma = 3a$ and $\varpi = \delta + \mathrm{reserve}$, with $2 \le Z^{\mathrm{reserve}}$. The source's `main_energy_common` proves smooth domination including the fixed support factor. The second transform accepts every positive radial scale. The extracted errors keep their original error radii and all three moving-character alternatives; no low-bin heredity is used in those branches.

G Assembly of the local physical moment

The purpose of this section is to compose the source identities, rather than to regard the favorable exponent ledger as an analytic estimate by itself. All references B3.· are to the pinned Cycle25 manuscript. We use its existing 9/2 moment and its unrestricted zero-slot moment, including their literal natural coefficients and eligibility conditions. Neither is replaced by the proposed stronger moment at any descendant state. The unmarked density is the fixed-dictionary theorem proved in `PlainUnmarkedDensity.tex`; its signed-row identity is supplied by `SIGNED_ROW_RECONSTRUCTION.md`, and the actual fixed dictionary, natural conductor, and literal eligibility transport are supplied by `FixedDictionaryBridge.tex`. Uniform finite Euler deletion and Fourier integration are supplied by `FourierMaskTransport.tex`.

Proposition G.1 (Fixed-family local physical moment)**.** *Assume the incumbent bound $\beta_* \le 7/8$, and work in its nontrivial contradiction branch $\beta_* \ge 51/100$. Fix an admissible natural parent family of B3.4. Its whole-product character is a fixed finite ray character times $\chi_n(k)$, with no parent moving support of positive width. Fix its ray modulus, anchor character, and nonzero fixed ideal. In particular retain the actual $Q_* = \mathrm{internalQ}(Q, \eta_0)$ and the literal exclusion $\neg\,\mathrm{FixedInducingRow}$ associated to that ideal; Q_* is proper, nonzero, and contained in* (72) *as in the source. Use the source's canonical prime pools, one common annular slot profile, and individual allowed relative ray characters. A common additional puncture of norm at most U^B is allowed. It is independent of the summed row, but may depend on frozen outer labels; the common Fourier majorant below is taken before those labels are summed.*

Write $U = Z^M$, $A = n_1 + n_2 + z$, and $C = 2249/500$. Suppose

$$\frac{41M}{100} \le n_i \le \frac{11M}{25}, \qquad \frac{3M}{100} \le z \le \frac{M}{25}, \qquad n_1 + n_2 + Cz \le M. \tag{LA.1}$$

For every requested $\epsilon > 0$, sufficiently fine canonical slots satisfy

$$\sum_{k \text{ eligible}} |S_{k,1}(Z^{n_1}) S_{k,2}(Z^{n_2}) Q_{k,I}|^2 \ll Z^{M+\epsilon} \mathcal{H}_J. \tag{LA.2}$$

The source radial majorant is understood. The factor $\mathcal{H}_J$ contains finite smooth seminorms and polynomial costs of the independent plain and slot heights. The same assertion holds for the four derivative profile pairs used by two-variable scale-supremum recovery.

The mesh can be selected from the requested loss and fixed bounded geometric ranges, before the eventual finite slot count and live labels. For the fixed arithmetic family and slot system, the finite profile and height orders, constants, and eventual threshold are selected before the live scales, masks, row selections, and heights. This is the fixed-family quantifier order proved here. No assertion that the compiled new density adapter already chooses its height order before every fixed character modulus is needed or made.

Proposed source proof, under review. One can normalize the argument to base U and $M = 1$; we retain M in the displayed ledgers to make their relation with the source transparent. Put

$$a = \frac{M}{2000}, \quad \sigma = 3a, \quad k_0 = \frac{7}{10}, \qquad g = (n_1 + n_2 + \tfrac{9}{2} z - M)_+.$$

The elementary consequences of (LA.1) are

$$g \le \frac{M}{12500} < \frac{M}{10000}, \qquad \frac{85M}{100} \le A \le \frac{44753M}{50000} < \frac{9M}{10}. \tag{LA.3}$$

All small positive losses denoted below by η are allocated from a single prescribed budget. That budget is smaller than a fixed fraction of both ϵ and $4M/225000$. The mesh is smaller than $a/10$ and a sufficiently small fraction of this budget. In particular one greedy overshoot fits this budget. After the finite slot list has been fixed, its aggregate support tolerance tends to zero as $Z \to \infty$; it is not charged once per slot.

1. Comparison with the same masks and rectangles. Let $L = M/4$. Both original plain lengths exceed L. Form the two equal-product rectangles with comparison lengths $L, n_1 + n_2 - L$, exactly as in B3.50. The whole-product character, the two profiles, the original slot coefficients, and every common puncture are identical in the two rectangles. This equality is retained through every later extraction.

To bound the comparison separately, expand only its additional common mask by B3.41. The absolute mass of the two divisor normalizers and the frozen prime-slot coefficients is $Z^{o(1)}$. This follows uniformly for polynomial-size masks from

$$\prod_{p|R}(1-(Np)^{-1/2})^{-1} \ll_\eta Z^\eta.$$

Every resulting first natural plain scale has length at most $M/4$; clipping a nonempty subunit scale to zero does not increase that bound. Reflect the second natural factor only if its length exceeds $M/2$. The combined conductor bound B3.42, its exact deletion/restoration identity B3.44, and the source scale-supremum argument put every retained reflected length below $M/2 + \xi$. The unreflected second factors are already below $M/2$. The same eligible rows remain nonprincipal for this reflection. Mask invariance preserves their literal exclusion; deleting slots only restricts the permitted positive family, or leaves the nonprincipal zero-slot family if all slots disappear.

Thus every comparison daughter has incumbent budget at most

$$M/4 + M/2 + \xi + (9/2)(M/25) = 93M/100 + \xi < M.$$

Use the existing $9/2$ moment, and sum the absolute coefficient masses and common-profile scale boxes. All independent heights remain in the profile controls. This proves the comparison estimate uniformly. The original centered difference still has its original whole-product mask; no indicator is inserted into a Poisson summation formula.

2. First transform and the joint first-label Cauchy step. Enlarge the positive norm of the whole centered difference to the source smooth row ball, as in B3.5. Its first zero frequency is bounded by $Z^{M+\eta}$ using the powerful-product count in B3.6. Apply the exact first transform to the remaining frequencies, including the complete first coprimality projector and its signs. Work in a fixed dyadic sector of the common widths c, d, p, R, E, s_0. Here p is the first common radical width and s_0 is the first projector width. Retain the common labels, including that projector, inside the Hilbert-space direct sum when applying Cauchy.

Let λ denote those labels and $\mathcal{N}_{C,\lambda}$, $\mathcal{N}_{D,\lambda}$ the exact positive Gauss norms B3.54. Set $B_c = (3c - 5d - R)_+/6$ and define B_d by exchange of sides. The outside power from the first transform is $Z^{M-A-R/2-E}$. It suffices to prove

$$\sum_\lambda \mathcal{N}_{C,\lambda} \ll Z^{p+A-c+R+E+B_c+\eta}, \qquad \sum_\lambda \mathcal{N}_{D,\lambda} \ll Z^{p+A-d+R+E+B_d+\eta}. \tag{LA.4}$$

Indeed joint Cauchy then gives total exponent

$$M + p - \frac{c+d}{2} + \frac{R}{2} + \frac{B_c + B_d}{2} + \eta \le M + \eta$$

by the primewise balance B3.56. No independent count of λ is made after (LA.4). For the branches treated per label below, the old allowance contains $-s_0$; the elementary first-label count $Z^{p+s_0+\eta}$ supplies exactly (LA.4). For the high branch the joint count will give (LA.4) directly.

The common Fourier measures are taken before the live labels, using the ambient-box construction of B3.3 and the transport supplement. If a continuous profile parameter varies with an outer label, use the common integrable Fourier-density majorant provided there; do not select a different uncontrolled Fourier measure inside each summand.

3. Amplification and the actual main radius. Use the auxiliary good primes of norm in $(Z^a/2, Z^a]$, disjoint from the live slots. Deleting primes in the row, common supports, masks, or first projector removes $O(\log Z)$ choices. The averaging and Gauss identities B3.58 give one main amplified norm with density $Z^{-a+\eta}$ and the three extracted error alternatives of B3.59. The normalized prime average in those errors remains normalized.

For the main norm use the genuine smooth ball with exponent

$$K_{\mathrm{main}} = K_0 + (d-c)_+ + 6a + \omega, \qquad K_0 = 2A - c - d + R + E - M. \tag{LA.5}$$

Here $\omega > 0$ includes the first retained-frequency padding and the fixed support-doubling reserve. Since the retained first dyad obeys $K \le K_0 + \delta_1$ and $\omega \ge \delta_1$, this ball covers the amplified range of exponent $K + 6a$, including its fixed support factors for sufficiently large Z. More precisely this is the existing `mainCommonRadius`, with $\sigma = 3a$ and $\omega = \delta +$ reserve; `main_energy_common` proves the majorization. Thus (LA.5) changes the actual smooth ball, not merely its declared upper bound. The Poisson identity B3.60 is valid at this ball radius. The extracted errors retain their old unamplified radii.

In the main second transform put $j = G_c V h'$, with the genuine common-support partition of B3.62–B3.63. Its nominal child width is

$$M' = M - (c-d)_+ - 6a - \omega - g_2 + t_2 \le M - 6a - \omega, \tag{LA.6}$$

since $g_2 \ge t_2$. For the C-side pre-allocation total $A' = A - c - c_2$,

$$(A' - M') - (A - M) = 6a + \omega - \min(c,d) - (c_2 - g_2) - t_2 \le 6a + \omega. \tag{LA.7}$$

These are direct substitutions for the actual radius; they do not require an unchanged old unpadded radius. The nominal inner allowance after the exact extraction normalizations is still $M' + \Delta$, where

$$\Delta = b_2 - p_2 + B_c + a \ge a. \tag{LA.8}$$

4. Literal children, pointwise gates, and low bins. Keep the full second projector. Its exact allocation B3.65–B3.69 leaves canonical slots with their original relative characters and the same whole-product character on each side. It does not insert new punctures on the unrestricted plain quotients. The first projector divides both genuine second common parts; the second projector has no such divisibility assertion.

The moving child character τ is distinct from the fixed slot anchor η_0. Its fixed exclusion ideal remains Q_*, not an ideal recomputed by inserting τ into the anchor. The source relative-character dictionary and `canonical_all_twists` show that each ray presentation actually used in the prime bounds is nonprincipal on canonical kept rows. The plain presentation is nonprincipal by the corresponding original row gate. Mask transport preserves both the exclusion and every positive-real zero maximum. Choose the finite dictionary of these presentations and the fixed reciprocity/unit sectors before live labels. Each zero bin is formed within one fixed parent translate $\tau_0 \operatorname{Char}(Q_*)$; different parent translates are partitioned first, at a fixed finite cost. Conjugation of a whole rectangle is conjugation of its whole character, whose zero maxima agree after height reflection. In particular $\overline{\eta\psi_w}$ must not be replaced by $\overline{\eta}\psi_w$. These identifications and the precise coset scope are proved in `FixedDictionaryBridge.tex`. Principal rows are handled in paragraph 7, not by a high-bin pointwise estimate. No arbitrary additional twist is inserted into this nonprincipal dictionary without its own gate.

Attach the common buffered bins to the actual effective row described in paragraph 5. A low bin has upper zero parameter at most $k_0 + O(e)$. The original pointwise prime estimate, applied first to the natural nonprincipal presentation and then transported across the polynomial-size Euler holes, bounds removal of slots of total width d_z by squared cost $Z^{(k_0+O(e))d_z+\eta}$. This applies to the actual canonical prime sums, including the original nonunit zeros.

The indicator of this low bin is imposed only after both transforms, on a positive child energy. The identities `AllocatedChildren.commonChild_energy`, `allocated_radial_bound`, and `AllocatedRayDictionary.common_clipped_energy` allow this arbitrary additional `keep` predicate. Pointwise remove one greedy prefix until the existing $9/2$ capacity holds, or until no slots remain. Then enlarge the remaining positive energy to the full canonical eligible family and use the incumbent moment. In the exhausted case use its unrestricted zero-slot estimate, which does not require the ordinary sum of the plain lengths to be below the width. No Poisson formula is applied to the low-bin indicator and no improved descendant moment is invoked.

There is a small but necessary width correction. Let d_* be the actual canonical width minus M'. Its upper bound is the declared support padding δ_2, but it can be negative if displayed moving support cancels. If F is the nominal affine defect, then the actual defect is at most $(F - d_*)_+$. For every $0 \le r \le 1$,

$$d_* + r(F - d_*)_+ \le \delta_2 + rF_+. \tag{LA.9}$$

This follows by considering the signs of $F - d_*$ and F; it is `width_plus_low_defect`. Apply it with $r = (k_0 + O(e))/(9/2)$. Equations (LA.7) and B3.67–B3.69 give the nominal defect bound $F \le g + 6a + \eta$. Consequently the combined width and slot-removal cost, including one greedy mesh increment, is at most

$$M' + \frac{7}{45}(g + 6a) + \eta.$$

It fits (LA.8), since

$$a - \frac{7}{45}(g + 6a) \ge \frac{M}{2000} - \frac{7}{45}\left(\frac{M}{10000} + \frac{3M}{1000}\right) = \frac{4M}{225000} > 0. \tag{LA.10}$$

Here ω, $O(e)$, the support padding, and the one mesh overshoot all belong to the already chosen smaller η budget. This proves the per-label low contribution within the old allowance; its first-label count gives (LA.4).

5. The main high bins are counted jointly. Use the signed local identities from the reconstruction supplement. For the relevant first side the actual row, including a possible fixed unit,

is

$$w = eF_cG_cVh', \qquad L_c = \log_Z NF_c, \qquad T_c = (c-d)_+.$$

The active residues in F_c are $(1+t_p) \bmod 6$ on the formal left and $(1-t_p) \bmod 6$ on the formal right. A zero residue retains its old puncture. Let R_0 be the canceled active radical width. The actual frequency support gives

$$\log_Z Nw \le W + O(\eta), \qquad W = M - R - T_c + L_c - 6a. \tag{LA.11}$$

The favorable extra $-\omega$ can be discarded here. Every genuine second common prime and the first projector divides w. The first common primes absent from w have total width at most $p-R-E+R_0$. At fixed w and their missing radical, all first and second common labels and the first projector are divisor-many, by the arbitrary-valuation reconstruction lemma. This is the actual signed source identity, not a guess based on a finite valuation table.

For $\delta \in [7/10, 3/4]$, put $\rho = 1 - 10\delta/[3(5-2\delta)]$. The proved unmarked density has a common scale threshold before height, and the sixth-power lifting is convergent because $\rho \ge 2/7 > 1/6$. It therefore counts these joint labels and rows by

$$Z^{\rho W + p - R - E + R_0 + \eta}(1+T_h)^J. \tag{LA.12}$$

A nonempty row range with slightly negative nominal W has $W \ge -O(\eta)$; apply density at the supported bound and charge this same tolerance. If the supported bound is negative, the range is empty. One does not use a population upper bound smaller than one for a nonempty bounded window.

The second projector label is not included in (LA.12). Allocate it exactly as in B3.65. Its dyadic count is paid by the two extraction scalars B3.69. The actual pointwise plain and slot estimates give paired normalized size at most $Z^{\delta(A-c-b_2)+\eta}$ before these reductions; each additional post-allocation reduction has a nonpositive contribution. Explicitly, B3.68 gives paired pointwise exponent $\delta(\alpha_1+\alpha_2-r_{\mathrm{clip},1}-r_{\mathrm{clip},2})/2 + O(\theta_N)$. B3.69 already pays the divisor dyad and both extraction scalars; the remaining $-\delta(r_{\mathrm{clip},1}+r_{\mathrm{clip},2})/2$ is nonpositive. This justifies the displayed normalized exponent without treating the second projector as a divisor of w. The outside power, from B3.60 with (LA.5), is

$$A - \min(c,d) + R + E - M + 5a + g_2 - t_2 - b_2 + O(\eta).$$

Relative to the joint target $p + A - c + R + E + B_c$, the resulting exponent is

$$\begin{aligned} &M(\rho-1) + \delta A + a(5-6\rho) \\ &\quad + (1-\rho)T_c - \delta c - (1+\rho)R - E - B_c + \rho L_c + R_0 \\ &\quad + g_2 - t_2 - (1+\delta)b_2 + O(\eta). \end{aligned} \tag{LA.13}$$

The second line is at most $(4\rho - 1 - \delta)p - E - B_c \le -79p/270 - E - B_c$: use $c = \min(c,d) + T_c$, $\min(c,d) \ge p \ge R$, $L_c \le 5(R-R_0)$, $\rho+\delta > 1$, and $5\rho > 1$. The third line without errors is at most $-\delta b_2 - t_2$. Finally (LA.3) implies the weaker bound $A/M \le 113/125$, and

$$1 - \rho - \delta A/M \ge 259/16875, \qquad 5 - 6\rho \le 23/7.$$

Thus, even allowing total high-branch errors $M/1000$, the first line and those errors are smaller than $-M/100$. Every bin straddling $7/10$ is assigned to the low branch, using its upper endpoint $7/10 + O(e)$ in the pointwise removal cost. The high branch contains only bins with lower endpoint at least $7/10$, so its zero-real-part parameter is at least $17/20$, exactly within the stated density range. The low-branch $O(e)$ rounding cost fits the reserved budget. This proves the main high part of (LA.4), directly with joint labels. There is no later Z^p or Z^{s_0} multiplication.

6. The extracted amplification errors. Retain the old error radii and use the fine form of B3.70. The three alternatives have

$$(w, w_o) = (\lambda, \lambda), (6\lambda, 0), (7\lambda, \lambda), \qquad \lambda \geq \sigma/4$$

for all sufficiently large Z. The nominal defect is at most $g + \sigma + w - w_o + \eta$, whereas the inner allowance has $\Delta \geq w$. Use global pointwise slot cost $3/4$ and incumbent capacity $9/2$, so the removal rate is $1/6$. The same width correction (LA.9) applies. The reserves are at least

$$13M/120000, \quad 193M/120000, \quad 119M/60000,$$

respectively, before small losses. These are obtained by substituting $g \leq M/10000$, $\sigma = 3M/2000$, and $\lambda \geq \sigma/4$. The moving factors at valuations one and seven and every extraction scalar are retained. The normalized prime average costs no second prime count. No low-bin heredity or new zero-density statement is used in these branches.

7. Zero frequencies and fixed-inducing children. The artificially added zero Gauss row is unamplified and has the direct B3.57 bound, including $-s_0$. The second zero frequency has, for (LA.5), excess

$$A - M - \min(c, d) - B_c + 5a + O(\eta) \leq A - M + 5a + O(\eta) < 0.$$

For fixed-inducing nonzero second rows, retain the paired exceptional predicate, forced congruence residues, raw extraction scalars, and common masks. B3.73–B3.79 apply with the actual smooth radius. For the main norm substitution into B3.75 gives exactly, before the ω padding,

$$F_1 = \begin{cases} (c + 5d + R + E)/6 + B_c - 4a, & c \geq d, \\ c + (R + E)/6 + B_c - 4a, & c \leq d. \end{cases}$$

The padding changes this by $-5\omega/6$. Thus $F_1 \geq 2c/3 - 4a - O(\eta)$ and $F_2 \geq 2b_2/3$. The earlier $-5a$ expression is a valid weaker lower bound, not the exact equality: it drops the positive amplifier density term $\ell = a$ in B3.75.

With $v = c + \min(c_2, d_2)$ and $L = M/4$, the exact maximum

$$\max_{v \geq 0} \{A - 5M/6 - 2v/3 - (L - v)_+\} = A - M$$

is attained at $v = L$. The main exceptional excess is consequently at most $A - M + 4a + O(\eta) < 0$. The unchanged error radii use the old bound $A - M + 3\sigma + O(\eta) < 0$. The mask-uniform lattice cancellation B3.77 cancels the two leading terms because the rectangles have the same product, character, mask, and whole-product norm power. Independent input heights are part of the two common profiles; they do not destroy that equality of leading terms. No exceptional contribution is hidden in an arbitrarily small error despite the fixed positive amplifier length.

8. Heights, losses, and recovery. The density theorem chooses its internal detector exponent first and returns a finite height degree. Choose a distinct external bin exponent $\tau_{\mathrm{out}} > 0$ only afterward, so every occurrence of $(1 + Z^{\tau_{\mathrm{out}}})^J$ fits the reserved power losses. For bounded input heights below a fixed small power of $Z^{\tau_{\mathrm{out}}}$, retain the separated frequencies in a still smaller part of that height buffer. All other common Fourier tails are paid with arbitrary integration-by-parts

orders, using raw polynomial bounds before any arithmetic enlargement. Those orders increase only the finite seminorm and allowed input-height powers.

Arbitrary larger input heights do not require a detector at an uncontrolled height: absolute counting gives a fixed polynomial bound in Z, and a sufficiently large fixed power of the largest input height absorbs it. This supplies the stated polynomial height allowance. The arithmetic scales have fixed bounded exponent ranges, so one such power suffices. The height threshold precedes the eventual choice of Z; it is not selected separately for each row.

There are only the two transforms and the calls to already established moments. Choose smooth derivative and height orders backward through this finite graph, then the Fourier-tail orders, and finally the eventual threshold. Coefficient divisor bounds and the finitely many dyadic sectors receive small shares of the original η budget. The first-label Cauchy step averages the two bounds (LA.4), and the inner Cauchy step averages its two child costs; neither duplicates a numerical loss. Minkowski adds coefficient masses already charged in the displayed estimates. The aggregate η losses are at most the requested ϵ.

Combining the comparison, zero-frequency, low, high, extracted-error, and exceptional estimates proves (LA.2). Replacing either plain profile by its logarithmic scale derivative repeats the same argument with finite increased seminorm orders. The source's two-variable Sobolev recovery therefore gives the four derivative-profile pairs and the stated bounded scale supremum. This proves the proposition. □

Scope of this assembly. This is an analytic source proof relative to the pinned incumbent theorems and the companion derivations explicitly named above. It is not a claim that a new complete Lean endpoint or any independent checker receipt exists. In particular the fixed-family quantifier order is stated explicitly: the proof supplies common exponent savings and a mesh chosen before live arithmetic labels, while constants and finite height orders may depend on the fixed arithmetic dictionary. To use this proposition in the endpoint argument, combine it with the exact endpoint certificate, the downward-scale dichotomy, the original weighted Euler coefficient mass, and the original target-uniform supremum contradiction. Those are separate from the local-moment estimate itself.

H Shortened physical scales do not require a new local hypothesis

The local moment window imposes a lower bound on each plain length. Consequently the sentence "the varying conductor only shortens the reflected scale" does not, by itself, justify applying that moment. The following dichotomy closes this particular gap. It uses the *existing* $9/2$ moment on the shorter scales.

Set

$$h = \frac{3969737987}{5000000000}, \qquad \ell = \frac{626701}{3750000}, \qquad y = h - 2\ell, \qquad C = \frac{2249}{500}.$$

For a row exponent $d \in [h - 10^{-3}, h + 10^{-10}]$ and a free-shift exponent $t \in [0, 10^{-3}]$, normalized relative to $U = Z^d$, the upper reflected plain length is

$$n(d,t) = 1 - \frac{y}{d} + t.$$

Since $y > 0$, this is increasing in both d and t. The exact endpoint values used in the scalar audit are

$$n_- = \frac{4998608000}{11894213961}, \qquad n_+ = \frac{402041377237}{952737117000},$$

and the nominal selected prime widths lie between

$$z_- = \frac{1-2n_+}{C} = \frac{74327181263}{2142705776133}, \qquad z_+ = \frac{1-2n_-}{C} = \frac{948498980500}{26750087198289}.$$

Direct rational comparison gives

$$\frac{41}{100} < n_- \le n_+ < \frac{211}{500} < \frac{11}{25}, \qquad \frac{3}{100} < z_- \le z_+ < \frac{71}{2000} < \frac{1}{25}.$$

All comparisons are strict, so sufficiently small fixed annular, box, fixed-conductor, and selection-mesh losses fit inside them. The numerical margins before these losses are approximately 0.0000143651 for the upper plain bound 211/500 and 0.0000422098 for the upper slot bound 71/2000. In particular those losses must be chosen below these margins; merely fitting inside the larger local window is insufficient for the quantitative short-scale reserve below.

Lemma H.1 (A downward scale dichotomy)**.** *Let $n, x_1, x_2, z \ge 0$ satisfy*

$$x_i \le n \le \frac{211}{500}, \qquad \frac{3}{100} \le z \le \frac{71}{2000}, \qquad 2n + Cz \le 1.$$

Then either both actual plain lengths x_i belong to $[41/100, 11/25]$ and

$$x_1 + x_2 + Cz \le 1,$$

or

$$x_1 + x_2 + \frac{9}{2}z \le 1 - \frac{33}{4000}.$$

Proof. If both $x_i \ge 41/100$, their upper bounds follow from $211/500 < 11/25$, and their new capacity follows by comparison with $2n + Cz$. If, for example, $x_1 \le 41/100$, then

$$x_1 + x_2 + \frac{9}{2}z \le \frac{41}{100} + \frac{211}{500} + \frac{9}{2}\frac{71}{2000} = 1 - \frac{33}{4000}.$$

The other case is symmetric. Equality at the boundary is harmless. □

Here is the precise analytic application. Assume the proposed local moment has been established for its stated profile and canonical-slot family, and retain the pinned source's existing 9/2 moment. Fix a selected slot family of total width z obeying the lemma, with the same natural row zeros and eligibility in both estimates. For any actual pair of reflected scales bounded above by U^n, clip nonempty subunit scales to length zero using the source's bounded annular dilation rule. Apply the local moment on the rectangle $x_1, x_2 \ge 41/100$; on either complementary rectangle apply the incumbent moment. Both bounds have the same target $U^{1+\epsilon}$.

The original scale-supremum recovery applies separately on these three rectangles. In logarithmic scale coordinates it bounds the supremum by integrals of the function and its first derivatives in each coordinate. These are exactly the four profile pairs appearing in the two-variable recovery. Their supports remain fixed annuli; the number of scale boxes is polynomial in $\log U$, and is absorbed in an arbitrarily small part of ϵ. Thus row-dependent conductor deficits, downward Euler-restoration scales, and the low annuli of the reflected Schwartz profile do not force a local-moment claim below 41/100. The upper annular tail is first removed with the original rapid-decay estimate.

The algebra relating this to the actual numerator is the pinned source's B0.42–B0.44: write a monomial shift as $h_0 h_r$, with h_r charged to the repeated-prime conductor deficit. Its reflected

scale is bounded by a fixed constant times $(U/Y)h_0$; natural Euler restoration only shortens that scale. For a fixed h_0 dyad this is the upper bound $n(d,t)$ above. The same combined conductor must be used in this bound and in the coefficient damping. No factor of the conductor deficit may be charged twice. The argument introduces no further coefficient mass and therefore preserves the source's weighted free-shift mass $\sum |c_u(h_0,h_r)|h_0^{\delta/2}$.

Short actual prime supply. There is a second possible shortening: the actual selected prime width z can be smaller than the nominal selection capacity $z_A = (1-2n)/C$. Error slots carry zero gain, and even a main slot may carry zero gain. No lower bound on the positive-gain supply is required. If $z \le 3/100$, the old moment applies directly, since

$$x_1 + x_2 + \frac{9}{2}z \le 2\frac{211}{500} + \frac{9}{2}\frac{3}{100} = 1 - \frac{21}{1000}.$$

Thus every actual selected width $0 \le z \le z_A \le 71/2000$ is covered: use this old-moment branch for $z \le 3/100$, and the preceding downward-scale dichotomy otherwise. The common worst reserve in either old-moment branch is $33/4000$.

The selection gain is still the gain at the *target capacity* z_A, not merely at the smaller actual width z. To see this from the literal source, write w_i for every original slot width, $L = \sum_i w_i$ for their total, and $g_i \in [0, \delta/2]$ for the main-masked amplitude-bin gains. The error gains are zero but their widths are retained in L. Set

$$q = \frac{\sum_i w_i g_i}{L}.$$

The source theorem `zero_error_whole_slots` assumes only $0 \le z_A \le L$, $w_i \le$ mesh, and the above gain bounds. It returns a subset I of the positive-gain main slots satisfying

$$z = \sum_{i\in I} w_i \le z_A, \qquad \sum_{i\in I} w_i g_i \ge q z_A - \frac{\delta}{2}\text{mesh}, \qquad \left|\prod_{i\in I} Q_i\right|^2 \ge U^{2qz_A - \delta\text{mesh}}.$$

There is no premise that the positive-gain main supply is at least z_A. If it is exhausted first, the selected weighted gain is $qL \ge qz_A$, which is stronger than the stated rounded bound. Zero-gain main factors must not be reinserted into this lower-bounded product; such factors could be arbitrarily small or zero.

The source's `source_class_main_selection` constructs the amplitude classes on all original slots and uses exactly this masked mean. Consequently the same I, q, and lower bound work uniformly on its class of rows. Its full-main *upper* bound is unchanged. To substitute the local capacity for its old capacity one invokes `zero_error_whole_slots` at z_A directly, after checking the total-original-supply inequality $z_A \le L = \ell/d$. This is the scalar audit's capacity gate; it does not assert a fictitious positive-main supply lower bound.

Combining this selection with either applicable physical moment gives the same denominator gain U^{2qz_A}, up to the fixed mesh loss, whether the actual width is in the local window or below it. The old-moment branch therefore preserves the desired weighted numerator saving. This argument supplies neither the new local moment nor its literal initial-state adapter; it closes the downward-scale and short-selected-supply parts conditional on those inputs and the pinned incumbent moment.

The six elementary statements in `ScaleShortening.lean` check the rational upper edges, both strict incumbent reserves, the two dichotomies, and the exhausted-supply affine gain. Their scope is arithmetic; the actual selection and spike identity are the source theorems identified above.

All larger free shifts, including zero capacity. For completeness the restriction $t \le 10^{-3}$ does not discard the remaining coefficient dyads. Put $b = 2y/d - 1 > 0$, $t_0 = 10^{-3}$, and

$$z_{\mathrm{new}}(0) = \frac{b}{C}, \qquad z_{\mathrm{old}}(t) = \max\left\{0, \frac{b - 2t}{9/2}\right\}.$$

The actual incumbent capacity uses $6\kappa \le 9/2$, so this is a lower admissible capacity. The weighted coefficient mass in B0.44 pays $U^{-\delta t/2}$. Thus it suffices to prove

$$\frac{\delta t}{2} + \frac{q z_{\mathrm{old}}(t)}{2} \ge \frac{q z_{\mathrm{new}}(0)}{2}. \tag{SS}$$

Write

$$S_b(t) = \frac{t}{2} - \frac{1}{4}(b/C - z_{\mathrm{old}}(t)).$$

This function is continuous at $t = b/2$. Its derivative is $7/18$ where the old capacity is positive and $1/2$ where it is zero. For the audited near-endpoint rows, $b \le b_{\max} = 1 - 2n_-$ and $b > 2t_0$. The exact rational audit gives

$$S_b(t_0) \ge \frac{t_0}{2} - \frac{1}{4}\left(\frac{b_{\max}}{C} - \frac{b_{\max} - 2t_0}{9/2}\right) > \frac{3}{10000}.$$

It follows that $S_b(t) > 0$ for every $t \ge t_0$, with no exception at the clipping point. Since $0 \le q \le \delta/2$ and $z_{\mathrm{new}}(0) - z_{\mathrm{old}}(t) \ge 0$, the left side of (SS) minus its right side is at least $\delta S_b(t) \ge 0$. This proves the full large-shift payment, including the empty selected family at zero capacity. That family uses the incumbent unrestricted zero-slot moment B3.40, not a moment with a hidden requirement $x_1 + x_2 \le 1$. In the source definition `ZeroAt`, the positive scales are required only to satisfy $X_i \le Z^L$; its conclusion is $Z^{s.\mathrm{width}+\epsilon}$. The actual physical specialization `finite_pair_zeroAt` in `Cycle25/Energy/Physical/CertifiedZero.lean` gives $U^{1+\delta+\epsilon}$ for $X_i \le U^2$. Its `certified_reflected_zero_fourier_moment` retains both independent Fourier heights. Finally `certified_original_numerator_fourth` in `Cycle25/Energy/Physical/OriginalFourth.lean` accepts the exact disjunction

$$2(m + \xi) + 6\kappa \sum_{i \in I} w_i \le 1 \quad \text{or} \quad I = \varnothing, \qquad m + \xi \le 2,$$

and gives the same $U^{1+o(1)}$ fourth-moment bound in either case. There is consequently no extra positive-part length cost when the capacity vanishes. The larger $\max(1, 2m)$ bound in the unmarked density adapter allows its finitely many exceptional rows; it is not the eligible physical zero-slot estimate being used here.

For $0 \le t \le t_0$ the local capacity is $z_{\mathrm{new}}(t) = (b - 2t)/C > 0$. Here the same conclusion follows directly from

$$\frac{\delta t}{2} + \frac{q}{2}(z_{\mathrm{new}}(t) - z_{\mathrm{new}}(0)) = t\left(\frac{\delta}{2} - \frac{q}{C}\right) \ge 0.$$

The original support cutoff restricts $t \le m + o(1)$, as in B0.44; all scales therefore stay in bounded geometric ranges. The new argument requires no new uniform estimate for unbounded plain lengths. Row exponents below the near-endpoint interval use the separate incumbent row-endpoint certificates, as specified in the global endpoint assembly.

I The endpoint and the target-uniform continuation argument

We now insert the local physical moment into the pinned numerator construction. The input results are the established incumbent moments, the exact physical Euler expansion and coefficient mass in B0.32–B0.44 and B4 of the pinned Cycle25 manuscript, its detector count B0.40, and its contour and continuation construction B0.7 and T2–T3. The new input is the local-moment proposition proved in `PhysicalLocalAssembly.tex`, with the fixed-dictionary and height-uniform density arguments preceding it. The downward-scale argument is `PhysicalScaleShortening.tex`. We retain the exact rational certificate in `local_moment_endpoint_audit.py`. No unproved varying-parent moment is introduced in this application.

Set

$$\theta = \frac{4374433}{5000000}, \quad \ell = \frac{626701}{3750000}, \quad h = \frac{3969737987}{5000000000}, \quad x = 1 + \ell - h, \quad y = h - 2\ell, \qquad C = \frac{2249}{500}.$$

In particular $\theta = 11/12 - \ell/4$, $x + y = 1 - \ell$, $0 < \ell < x < y < 2x$, and $y < 1/2 < h < 2y$.

Theorem I.1 (Proposed endpoint, relative to the pinned analytic framework)**.** *Every finite-order Hecke character of the Eisenstein field with trivial infinite type has no zero in $\Re s > \theta$, with the principal pole at $s = 1$ excluded. Presented characters may be imprimitive and have arbitrary conductor. Every ordinary Dirichlet L-function has the same zero-free half-plane, again excluding the principal pole. In particular $\zeta(s) \neq 0$ there for $s \neq 1$.*

Proposed source proof, under review. Let β_* be the source's Hecke zero supremum, including its sentinel $1/2$. The incumbent theorem gives $\beta_* \le 7/8$. Suppose $\mu = \beta_* - \theta > 0$ and set

$$\kappa = 2\beta_* - 1, \qquad \kappa_* = 2\theta - 1 = \frac{1874433}{2500000}.$$

Then $\kappa_* < \kappa \le 3/4$. The incumbent marked detector and ordinary moment apply at this κ: their hypothesis $2\beta_* - 1 \le \kappa$ holds by equality. The new local moment uses only the incumbent $9/2$ and unrestricted zero-slot estimates at descendant states.

1. The actual physical family and selected slots. Fix a primitive target η only after the common real margins and slot mesh have been chosen. Its finite exclusion set S, original local Euler factors, and all natural row zeros are those of the source probe. In the physical numerator moment the fixed base is `fixedSourcePrincipal` S; the variable part is the sextic row character. This is exactly the coefficient identity in `PhysicalMomentInput.source_physical_slot_conj` and `reflected_pair_norm_eq`. The relative prime coefficient contains the inverse fixed base and the common conjugate annular profile, with real power $33/50$ and individual height $-\Im z_i$. Conjugation applies to the entire reflected product. Thus the local proposition applies to this fixed family with the same literal eligibility and masks. The fixed exceptional free rows have bounded norm and are absent from every growing retained dyad; the source `weighted_source_exceptional_norm` proves this fact.

In an amplitude class let $w_i = \ell_i/d$, let $L = \sum_i w_i = \ell/d$, and let $0 \le g_i \le \delta/2$ be its amplitude exponents. Error slots have $g_i = 0$. Put $q = L^{-1}\sum_i w_i g_i = \delta p$, where $0 \le p \le 1/2$. For any target capacity $0 \le z_A \le L$, the exact theorem `SlotSelection.zero_error_whole_slots` selects positive-gain main slots I such that

$$z_I = \sum_{i \in I} w_i \le z_A, \qquad \left|\prod_{i \in I} Q_i\right|^2 \ge U^{2qz_A - \delta\,\mathrm{mesh} - o(1)}. \tag{EA.1}$$

The gain is measured at the target capacity z_A, even if z_I is strictly smaller. It is not legitimate to assume z_I nearly fills z_A or to insert error slots into the selected product. The selection is made once for the amplitude class, not independently for its rows.

Put $d_0 = 1/200$ and $\zeta_0 = 10^{-10}$. All original slots can be fixed so short that their relative widths meet every required mesh for $d \geq d_0$. Detector supply is available because the exact geometry satisfies $\ell/(h + \zeta_0) > 4/21$. For $d < 1/2$, the same selection theorem applies directly, without the $d \geq 1/2$ assumption of its convenience wrapper: $L \geq 2\ell > 1/3$ and the old physical capacity is at most $1/(6\kappa) \leq 5/21$. Thus no lower-row supply condition is missing.

2. The reflected scale and every free Euler shift. The exact coefficient expansion writes a shift as $h_0 h_r$, with h_r charged to the repeated-prime conductor deficit. B0.42–B0.44 give the actual reflected scale bound

$$X_u \ll_S (U/Y)h_0, \qquad \sum_{h_0,h_r} |c_u(h_0,h_r)|h_0^{\delta/2} \ll U^{\varepsilon}. \tag{EA.2}$$

The same combined conductor occurs in both statements. Its deficit is not charged a second time. Natural Euler restoration only shortens scales; its coefficients have the source's bounded absolute mass. The original support retains $h_0 \ll Y$, so all scale exponents lie in a fixed bounded range.

Consider first $h - 10^{-3} \leq d \leq h + \zeta_0$ and $t = \log_U h_0 \in [0, 10^{-3}]$. The upper reflected length is $n(d,t) = 1 - y/d + t$. Choose the target capacity just below

$$z_A(t) = \frac{1 - 2n(d,t)}{C},$$

reserving the small capacity decrement for annular and scale-box errors. The exact geometric checks, with strict room for those errors, give

$$\frac{41}{100} < n(d,t) \leq \frac{211}{500}, \qquad \frac{3}{100} < z_A(t) \leq \frac{71}{2000}, \qquad z_A(t) < L.$$

Let $x_1, x_2 \leq n(d,t)$ be the actual nonnegative plain lengths, and let z_I be the actual selected width from (EA.1). There are three cases. If $z_I < 3/100$, the incumbent capacity holds with strict reserve:

$$x_1 + x_2 + \frac{9}{2}z_I \leq 2\frac{211}{500} + \frac{9}{2}\frac{3}{100} = \frac{979}{1000} < 1. \tag{EA.3}$$

If $z_I \geq 3/100$ and some $x_j < 41/100$, the downward-scale lemma gives the incumbent reserve $33/4000$. In the remaining case the local proposition applies because $x_1 + x_2 + Cz_I \leq 2n(d,t) + Cz_A(t) \leq 1$. The empty selected product uses the unrestricted zero-slot estimate. The same two-variable scale-supremum recovery and its four derivative profiles apply on these separate regions. Therefore the fourth moment needed for numerator Hölder has the same $U^{1+\varepsilon}$ bound in all cases, while its gain remains $qz_A(t)/2$ by (EA.1).

For these small shifts,

$$-\frac{\delta t}{2} - \frac{qz_A(t)}{2} \leq -\frac{qz_A(0)}{2},$$

since $q \leq \delta/2$ and $C > 1$. For larger shifts use the incumbent capacity instead. Write $b = 2y/d - 1$ and use the smaller comparison capacity $z_o(t) = \max(0, (b - 2t)/(9/2))$. The exact scalar audit shows

$$S(10^{-3}) > \frac{3}{10000}, \qquad S(t) = \frac{t}{2} - \frac{b/C - z_o(t)}{4}.$$

While $z_o(t) > 0$, its derivative is $7/18$; after the capacity clips to zero the derivative is $1/2$. Hence $S(t) \geq 0$ for every $t \geq 10^{-3}$. With $q \leq \delta/2$ this proves

$$-\frac{\delta t}{2} - \frac{q z_o(t)}{2} \leq -\frac{q z_A(0)}{2}. \tag{EA.4}$$

The actual old capacity at coefficient $6\kappa \leq 9/2$ is no smaller. Thus (EA.2) pays every free-shift dyad, including those with zero old capacity. No Euler term has been omitted from the numerator inversion. At zero capacity the two reflected plain lengths can have sum greater than one; this is exactly where the *unrestricted* physical MP0 is needed. Its actual source adapter `Physical/CertifiedZero.finite_pair_zeroAt` has no affine plain-length condition and bounds the moment by $U^{1+\varepsilon}$ for each plain scale up to U^2. The theorem `Physical/OriginalFourth.certified_original_numerator_fourth` accordingly accepts either its affine capacity inequality or an empty selected family. The original support bound in (EA.2) gives reflected length at most $1 + o(1) < 2$, so this is a literal application of that theorem. The weaker all-free-row plain estimate used in the new density argument is not being substituted for this physical MP0. Below $d = h - 10^{-3}$ the original B0.41 argument is unchanged.

3. The common retained-row saving. The detector remains

$$R_\kappa(\delta, p) = 1 - \delta + \frac{\delta c_\kappa(p)(5/6 - \delta)}{2(5/6 - \delta + \delta c_\kappa(p))}, \qquad c_\kappa(p) = \frac{2(1-p)(1 - p/(3\kappa))}{3 - p - 2p/(3\kappa)}.$$

All $\delta \in [0, 3/4]$ and $p \in [0, 1/2]$ are included. The physical Hölder argument and the weighted mass (EA.2), on the original full w-inversion line, give the exponent $G_{C,\kappa}$ of the paper on the local row interval. On old rows its general form is

$$G_o(d) = \frac{1+\delta}{2} + h\frac{13}{75} - \frac{y}{2} - \frac{\ell}{2} + \ell q + d\left(\frac{1}{4} + \frac{3R_\kappa}{4} - \frac{17}{50}\right) - \frac{q}{12\kappa}\min(d, 2y - d). \tag{EA.5}$$

This is the exact B0.46/B0.55 ledger, retaining the reciprocal only after the full physical numerator inversion.

At $\kappa = \kappa_*$ the rational Bernstein certificate proves savings 10^{-7} at $(d, c) = (h, C)$, 10^{-5} at $(h - 10^{-3}, 6\kappa_*)$, and 10^{-3} at $(1/2, 6\kappa_*)$, uniformly on the full (δ, p) rectangle. At the junction the new exponent is no larger than the old one because $C < 6\kappa_*$ and $2y - d > 0$. Affinity in d covers the intervening local interval. The extension above h costs at most $2\zeta_0$, so the retained saving there is at least $499/5000000000 > 9 \cdot 10^{-8}$.

The old function (EA.5) is convex in d. To cover $d_0 \leq d \leq 1/2$, combine the certificate at $1/2$ with the elementary bound at d_0, using $a \leq 7/8$, $R_\kappa \leq 1$ and $q \leq 3/8$:

$$G_o(d_0) \leq \frac{7}{8} - \frac{49h}{150} + \frac{7\ell}{8} + \frac{33 d_0}{50} = \frac{573880513637}{750000000000} < \theta - \frac{1}{10}.$$

On $[1/2, h - 10^{-3}]$, affinity and its two certified endpoints suffice. The floor classes use the source bound

$$\frac{51}{100} + \frac{h}{3} + \frac{51\ell}{100} = \frac{12898208027}{15000000000} < \theta - \frac{1}{100},$$

with room for the tiny row extension. They do not require a positive detector zero above the sentinel.

The variation in κ must be checked separately for the two physical capacities. On the local branch only R_κ varies, and the source derivatives $\partial_\kappa c_\kappa \leq 1/18$ and $\partial_c R \leq 5/12$ give $\partial_\kappa G_{C,\kappa} < 1/50$.

Because $\kappa-\kappa_*=2\mu$, its increase is less than $\mu/25$. Thus its saving below β_* is at least its certificate saving plus $24\mu/25$. For the old branch the reciprocal capacity also varies. Since $\min(d,2y-d)\le 1/2$ and $d\le 1$, one has

$$\partial_\kappa G_o \le \frac{5}{288}+\frac{(3/8)(1/2)}{12(7/10)^2}=\frac{695}{14112}<\frac{1}{8}.$$

Its feedback saving is therefore at least its certificate saving plus $3\mu/4$. In particular all retained classes have a common strictly positive power reserve before a target character is fixed.

4. Principal, very small, very large, and off-height terms. The new geometry leaves the source contour boxes inside their stated domains; in particular $\theta>87/100$ and the physical scalar boundary is still exactly $11/12-\ell/4$. The principal double residue B0.35–B0.36 is the same function f_η. Its three remainders have relative reserves $y/4$, $h/24$, and a fixed positive prime-product exponent μ_P. Their derivation uses the local Euler identity and the principal correction, not the new moment.

For nonprincipal rows below Z^{d_0} the original global-line bound B0.56 gives, before small losses,

$$h\frac{13}{75}-\frac{y}{2}+\frac{63d_0}{50}=-\frac{21483987121}{250000000000}<-\frac{1}{20}.$$

The finite nonprincipal unit rows are included. For rows above $Z^{h+\zeta_0}$ the raw absolute estimate B0.57 applies: after choosing $\zeta_0>0$, choose its fixed positive z-line parameter v large enough that the term has any prescribed negative power. This is not an application of a central-row density estimate.

The compensated principal probe satisfies the actual B0.27 geometry: with $M=x+y=1-\ell$ one has

$$y-2x+\ell=-\frac{597590039}{5000000000}<0,\qquad 1+3\ell-2M=-\frac{123299}{750000}<0.$$

Thus its exponent is $(5x+y)/12=\theta-k$, where $k=5/6-x/3-\ell/6$. Division by $c_S A(Z)$ costs only a power of $\log Z$, giving $J_\eta(Z)\ll_\eta Z^{\theta-k+\varepsilon}$. The three independent contour frequencies and their joining segments are those of B0.7. Raw polynomial bounds are taken before any separation in a tail; they give $Z^{B'}T^{-N}$ with B' fixed before the tail order N. No contour crosses an unaccounted nonprincipal numerator pole or a reciprocal zero.

5. Order of choices and continuation. Choose from μ, the geometric and certified margins a common moment loss, capacity decrement, amplitude mesh, and a finite slot system meeting both the old and new moment meshes on $d\ge d_0$. Its smallest slot fixes $\mu_P>0$. Choose common real buffers smaller than these reserves. These choices, including the power saving below, precede the target η.

For each target now fix its finite set S, dictionary, and the finitely many internal moment, profile and Sobolev orders. They may depend on the target. Next choose a target-dependent positive analysis-height exponent τ small enough that all factors $(1+Z^\tau)^J$ use less than the same reserved real margin. The internal detector exponent in the density proof was already fixed; it is not this later τ. Choose the external tail order after τ, and finally the target-dependent asymptotic threshold. The resulting constants and thresholds may depend on the target, but one obtains a single $\Delta>0$, independent of it, with

$$J_\eta(Z)-f_\eta(Z)\ll_\eta Z^{\beta_*-k-\Delta}. \tag{EA.6}$$

Here Δ is smaller than fixed fractions of μ, the retained saving, $y/4, h/24, \mu_P$, and the small-row saving. Make it also smaller than $\mu/4$. The principal probe bound then gives $f_\eta(Z) \ll_\eta Z^{\beta_*-k-\Delta}$.

At $Z \downarrow 0$ the defining Mellin line can be moved arbitrarily far right, as in B0.60. Therefore

$$\int_0^\infty f_\eta(Z) Z^{-s+k} \frac{dZ}{Z}$$

is holomorphic for $\Re s > \beta_* - \Delta$. On $\Re s > 1$ its exact value is

$$e^{(s-5/6)^2} \frac{H_\eta(s)}{L^S(s,\eta)}.$$

The source cutoff ensures $|H_\eta(s) - 1| \le 1/2$ for $\Re s > \theta > 87/100$, and the Gaussian never vanishes. Since $\beta_* - \Delta > \theta$, holomorphic continuation excludes every zero of $L^S(s,\eta)$ to the right of $\beta_* - \Delta$, away from the principal pole. Restoring fixed Euler factors does not add zeros in this positive half-plane.

This is precisely the quantifier pattern of the proved, variable-boundary `Cycle25.PrimitiveProbeContract` and `UniformCommonProbe` in `Cycle25/Assembly/Continuation.lean`. In that file `beta_le_of_common_probe` first chooses a primitive zero sufficiently near the supremum and then invokes the target's individual probe. Its Big-O constants and threshold occur after the target quantifier. Thus no uniform numerical bound on growing target moduli is needed. A target-dependent power saving would be insufficient; (EA.6) has explicitly avoided that error. The common positive improvement contradicts the definition of β_*. Hence $\beta_* \le \theta$.

6. Primitive reduction and Dirichlet transfer. For an imprimitive Hecke character induced by η^*, the exact identity T2.1 expresses its L-function as $L(s,\eta^*)$ times finitely many factors $1 - \eta^*(\mathfrak{p})(N\mathfrak{p})^{-s}$. They are nonzero for $\Re s > 0$. This proves the Hecke assertion for every presented finite-order character of trivial infinite type, with the same pole exception.

For an ordinary Dirichlet character χ, use its norm lift to the Eisenstein field. The pinned theorem `HeckeDirichlet.LFunction_eq_dirichlet_product` gives

$$L_K(s,\eta_\chi) = L(s,\chi)L(s,\chi\chi_{-3})$$

away from $s = 0, 1$, with the original imprimitive and ramified Euler factors included. For $\theta < \Re s < 1$, both factors have finite values and their product is nonzero by the Hecke result, so $L(s,\chi) \ne 0$. On $\Re s \ge 1$ the classical ordinary Dirichlet nonvanishing theorem applies with its principal-pole exception. The trivial character of modulus one gives the stated zeta assertion. These are exactly the transfers in `Cycle25.dirichlet_of_hecke` and `zeta_of_hecke`; they require neither primitivity nor reality of χ. □

Verification status. This is a complete proposed written source-level assembly, under review, relative to the pinned incumbent analytic framework and the new local proof. The strict rational comparisons and Bernstein certificate are exact. Selected supporting Lean modules compile, but this file does not create a new all-family Lean theorem, a clean dependency replay, a canonical-target comparison, or an independent kernel certificate.

References

[1] Hailey Collet, *The quartic QRH boundary*, 10 October 2026. Cycle25 manuscript and source, revision `58344dfdbe756cf2f743da1908ffe0582418cf5b`. Pinned manuscript. Equation references B0 and B3 in the companion notes refer to this revision.

[2] OpenAI, *OpenAI/math*, formal mathematical development, revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Pinned source repository.

[3] Akash Levy, *Weighted-numerator proof*, source package, revision `2fd60c0926b66ea18d7436f5ed55250fd006ab5d`. Pinned proof package.

[4] Zhongpai Gao, with AI assistance, *Local physical moment research candidate: 0.8748866*, 11 October 2026. Source-review snapshot, revision `7f9166e2d626586cc3c89853497c0e0e22e856ee`. Research notes, certificates, and Lean components; draft pull request 15.

[5] Zhongpai Gao, with AI assistance, *Joint-label reconstruction and actual unmarked inputs*, 11 October 2026, second research revision. Companion note and nine compiled components. That historical revision kept the physical factorization as an application obligation; Draft 3 supplies the source-level arguments in the present supplements.